User manual for the SBB RCM-DX Viewer

Author	Gsponer Mischa (I-VU-UEW-MUD-DAM)
Status	In progress
Version	1.0
RCM-DX version	2.9.0
Last Modified	02.05.2025
Last amended by	Gsponer Mischa (I-VU-UEW-MUD-DAM)
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Filing	https://sbb.sharepoint.com/sites/mud/Monitoring/03_Monitoringdaten Prepare and Deploy/RCMDM/08_Projects/RCM OSS/User Guide OpenSource RCM.docx

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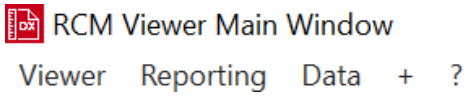
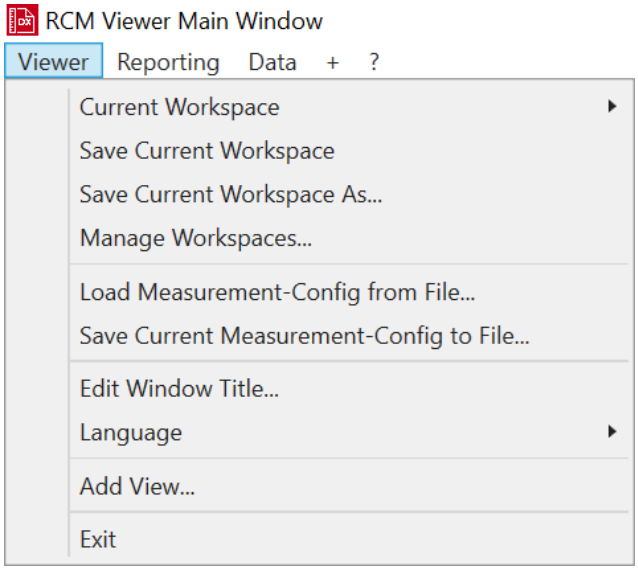
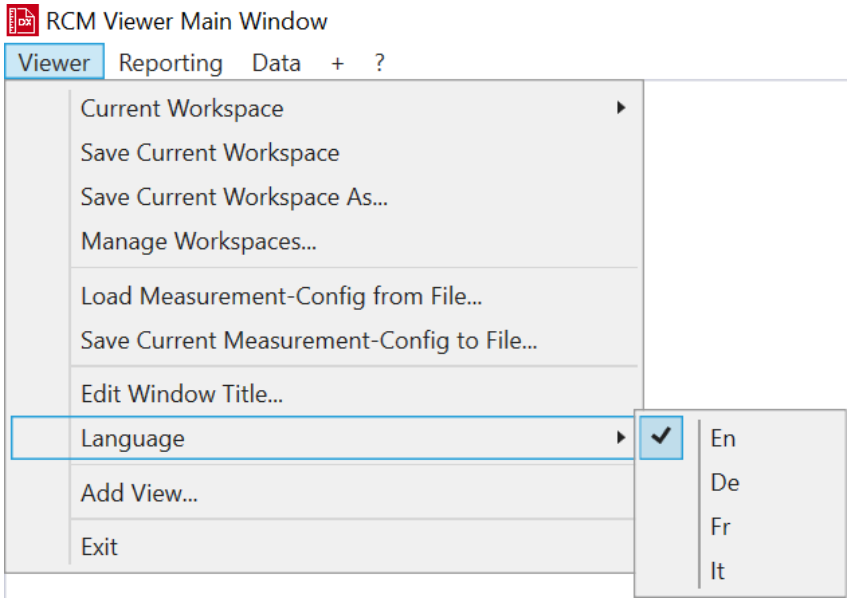
1. RCM-DX Viewer

Data collected from diagnostic vehicles and stored in the RCM-DX format can be visualised with the RCM-DX Viewer.
The application runs independently and can be installed on any PC/laptop with a Windows operating system.
The RCM-DX Viewer as freeware and the RCM-DX format under the open-source licence are made available by SBB free to use.
In the following sections, we will show the possibilities offered by the RCM-DX Viewer and how to work with the tool.

1.1. Menu bar



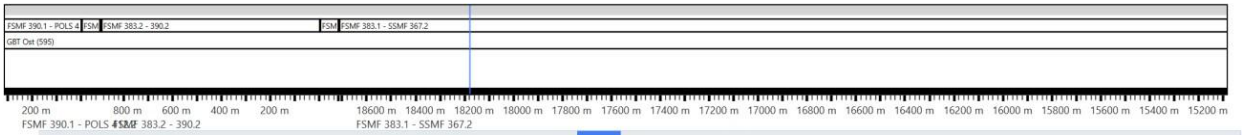




1.2. Template: Navigation bar

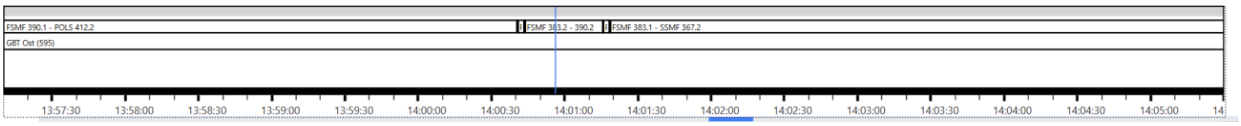
The navigation bar is used to orient and visualise the measuring section in the RCM-DX file. It contains different information depending on the mode (position mode or time/distance mode). How to switch the mode and what the differences are is described in Chapter 1.5 (Display Types/Modes).
The blue vertical line always shows the current position. All views display their information exactly at this point. The line can be moved with the mouse or you can jump directly to the desired position with a mouse click. However, the line can only be moved within the displayed area.
The horizontal bar at the bottom shows the current position in relation to the entire measuring distance. The bar can also be moved with the mouse and moved from the beginning to the end of the distance.
If the vertical line is no longer visible by moving the horizontal bar, you may need to move the current position to the desired position with a mouse click so that the information in the views is updated.

Position mode



In this mode, the GTG (Gleistopografie/Track topography) strands and Line are displayed. The specification of the current position is displayed in metres on the GTG strand. The metering starts again from 0 for each GTG strand and is displayed in ascending or descending order. This depends on the definition in the track topology and the direction of travel, as well as the vehicle orientation.
The direction of travel and vehicle orientation can be changed manually by the user and is described in Section 1.5.2 (Position mode).

Time/distance mode



In this mode, the GTG strands and Line are also displayed. However, the indication of the current position (blue vertical line) is displayed as the time or distance travelled over the diagnostic vehicle. The display is always in the direction of travel.
How to switch between time mode and distance mode is described in Chapter 1.5.1 (Time/distance mode).

1.3. Navigation functions

The navigation bar is supported or supplemented by the following functionalities. In addition to the automatic playback of the measuring section, navigation and zooming, other useful information is also displayed. This is described in more detail in the following sections.

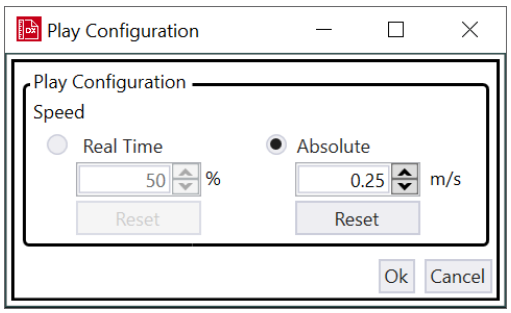
This icon can be used to show and hide the navigation bar and the navigation functions.

1.3.1. Play

The play function can be used to play the measurement run automatically. The position and the information in the individual views are automatically adjusted. The play and pause icon can be used to start and stop the playback. The desired speed can be set in the configuration (play icon with gearwheel).

1.3.1.1 Configuration (Play)

If the configuration icon is clicked, the following dialog opens:



Here you can choose between two types of playback speeds:

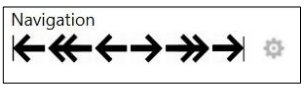
a) Real time:
   Percentage of speed driven in real time. 100% real time, 50% half speed or 150% 1.5 times speed.

b) Absolute:
Here, the playback speed can be specified absolute in m/s.

Press ‘Reset’ to reset the default value.

1.3.2. Navigation

The navigation arrows can be used to change the current position.



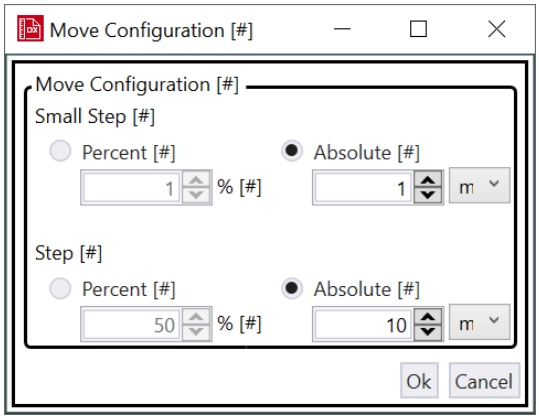
From left to right, these are the following functions:

- Jump to the start
- Step towards the start
- Small step towards the start
- Small step towards the end
- Step towards the end
- Jump to the end

The steps and small steps can be set with the gear wheel. More on this in the next chapter.

1.3.2.1 Configuration (navigation)

If the configuration icon is clicked, the following dialog opens:



There are two types for each step (step) and small step (small step):

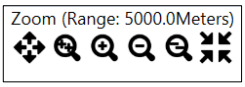
a) Percent:
Here you can enter the percentage of the total distance.

b) Absolute:
Here you can specify the steps in metres or seconds. You can select between m (meters) and s (seconds) in the dropdown.

These settings are retained even after closing the RCM-DX Viewer and do not need to be reset every time.

1.3.3. Zoom

With these buttons you can change the zoom level and thus the displayed area in the navigation bar. Via the buttons the currently displayed area (range) is displayed.



From left to right, these are the following functions:

- Full zoom in
- Step zoom in
- Small step zoom in
- Small step zoom out
- Step zoom out
- Full zoom out

The size of the steps is predefined and cannot be adjusted.

1.3.4. Position info

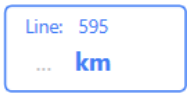
Information about the current position is displayed in the large blue box in the middle.

The following information can be seen on it:

- Time: Date and time of recording at the current position
- Distance: Rolled distance from start to current position
- Line: Line and kilometres at current position
- Track: GTG designation and distance in meters on the GTG to the current position

1.3.5. Line/km

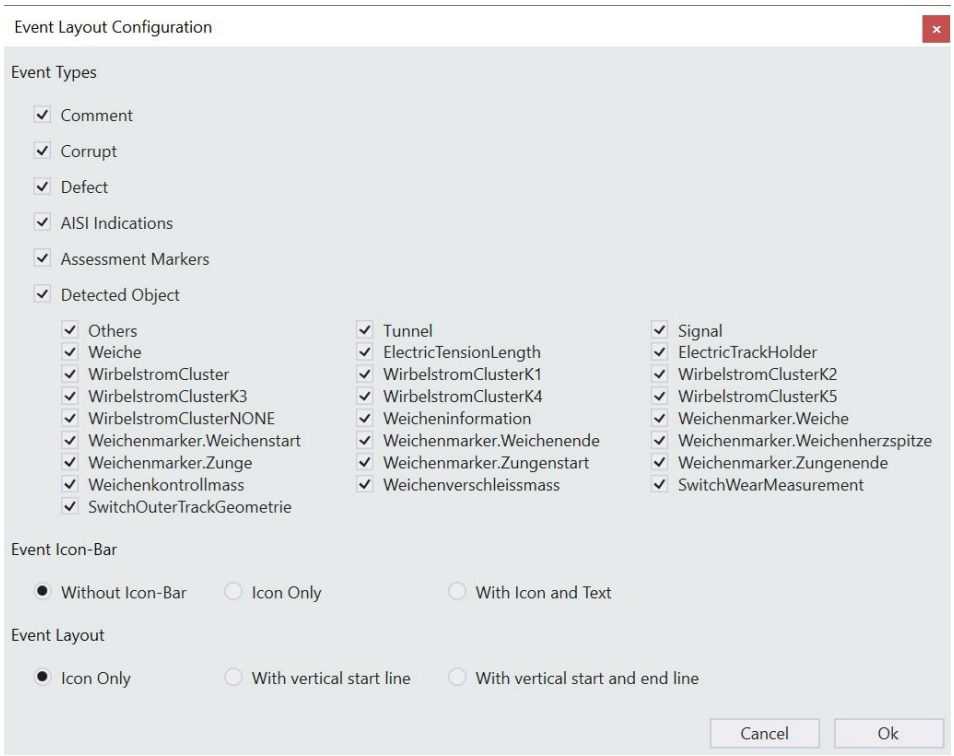
This blue box can be used to jump to a specific position. Enter the exact kilometre of the line shown above and press Enter to confirm.



The kilometre entered must be present on the measured route – otherwise you cannot jump. And if the kilometre you are looking for is on a different line than the one shown above, you must first change to the desired line with the help of the navigation and only then can the kilometre be entered.

1.4. Event representation configuration

The icon takes the user to the configuration view for the navigation area. Here can be set what should be displayed in the navigation area.



Type of events (event Types)
Setting of which event types are displayed in the navigation area. This view depends on the event types in the RCM-DX file.

Toolbar (event Icon Bar)
This is where the appearance of the toolbar is defined. The user can choose between:

- Hide toolbar
- Show only icons
- Show icons and text

Display type (event layout)
This is where the appearance of the events is defined. The user can choose between:

- Icon only
- With vertical starting line
- With vertical start and end line

1.5. Display types/modes

The RCM-DX Viewer distinguishes between two display types. Time/distance mode, in which the measurement run is displayed according to elapsed time or distance. Position mode, in which the measurement run is displayed according to the position on the line/km and GTG strand.

1.5.1. Time/distance mode

In time/distance mode, you can switch between time and distance display.

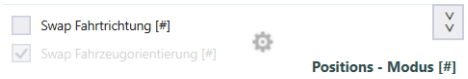
1.5.1.1 By time

In the navigation bar, the measurement run is displayed according to the elapsed time.

1.5.1.2 After the distance has been rolled out

In the navigation bar, the measurement run is displayed according to the rolled distance.

1.5.2. Position mode



1.5.2.1 Reverse travel direction

Reversing the direction of travel of the reference measurement run. This function is used for analysis views which show objects in a specific direction of travel.

1.5.2.2 Reversing vehicle orientation

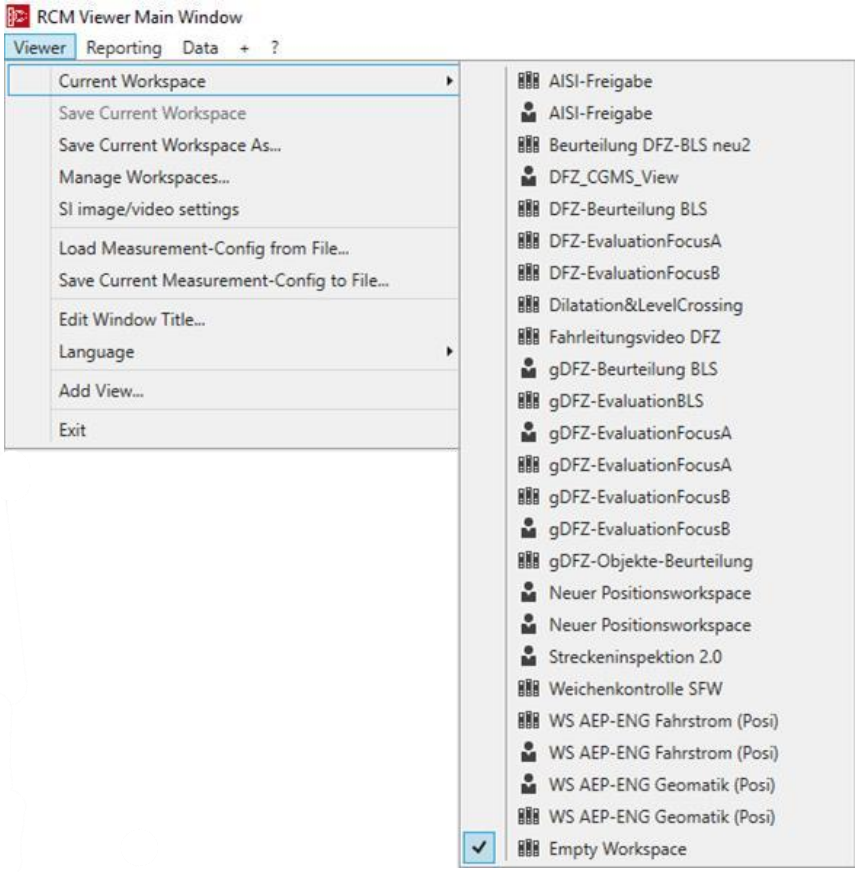
Reversing the vehicle orientation during the reference measurement run. This function is used for analysis views which correctly represent the left and right measurement channels. This option cannot be selected and is set automatically when the vehicle is reversed.

2. Workspaces

A workspace is a collection of views (see Chapter 6 - Views) that the user can individually compile, arrange and dimension within a workspace as desired. Any number of workspaces can be created and saved.
However, the information which data has been loaded is not stored in the workspace. After loading a workspace, the desired measurement data must always be searched for and loaded (see chapter 4.1 Loading measurement file). If the loaded data for a workspace are also to be stored, a new measurement configuration can be created (see chapter 3 Measurement configuration).

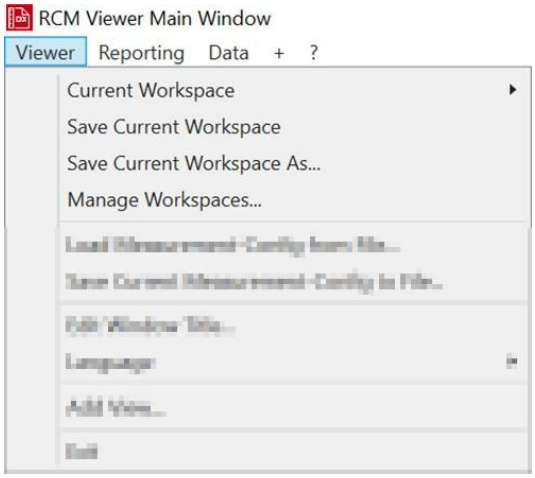
2.1. Select workspace (Current Workspace)

A saved workspace can be selected in the Viewer menu → Current Workspace. As soon as the workspace has been selected, the current workspace is closed and the new workspace is opened with all views and windows. The loaded data remains the same and is displayed in the new workspace.

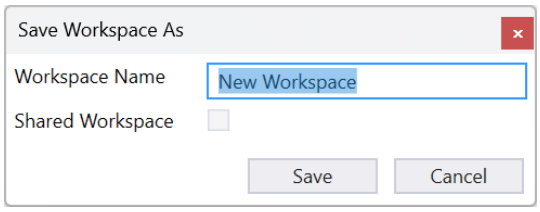


2.2. Save workspace

A new workspace or a changed workspace can be saved in the RCM-DX Viewer. ‘Save Current Workspace’ saves the changes to the existing workspace. ‘Save Current Worksapce As...’ saves a new workspace.



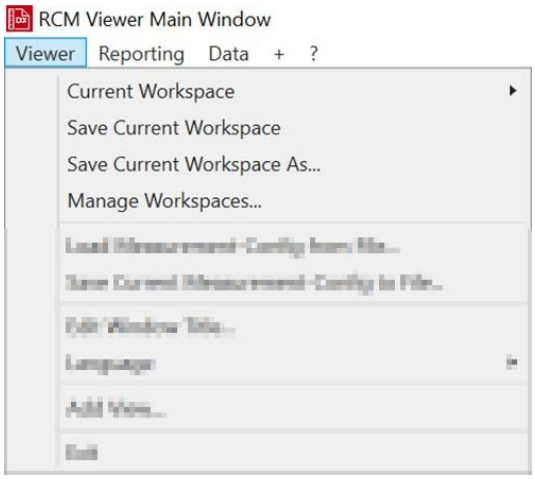
If a new workspace is saved, a dialog opens in which you can assign a name for the workspace.



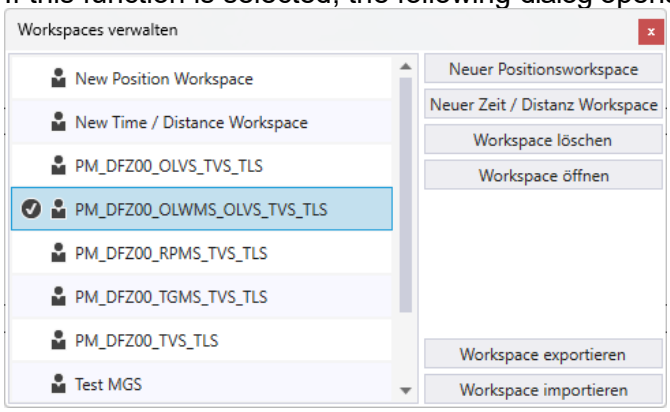
The ‘Shared Workspace’ checkbox is not activated in the RCM-DX Viewer. This is only available in the online version of the RCM Viewer.

2.3. Manage workspaces

The saved workspaces can be managed in the Viewer menu → Manage Workspaces.



If this function is selected, the following dialog opens:



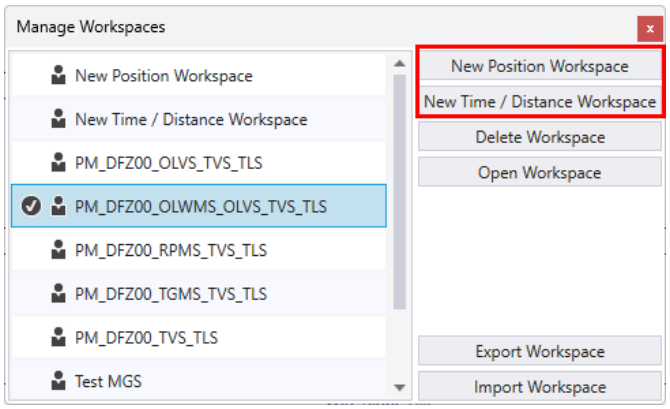
The saved workspaces can be managed here. The corresponding functions are described in more detail in the following sections.

Explanation of symbols:

- The currently loaded workspace
- User-specific workspace: The workspace is only available to the current user
- Shared workspace: The workspace is available to all users (only in the online version of the RCM viewer)

2.3.1. Create workspace

With the ‘New Position Workspace’ and ‘New Time / Distance Workspace’ functions, a new workspace can be created in the desired mode (‘Position’ or ‘Time / Distance’).

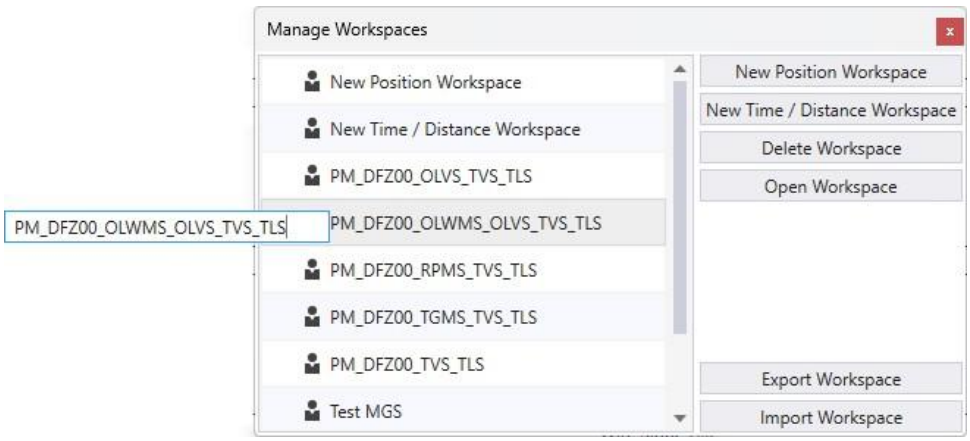


After the selection, a new, empty workspace is created in the corresponding mode, to which the desired views can then be added.

The two modes ‘Position’ and ‘Time / Distance’ are described in more detail in Section 1.5 Display types/modes.

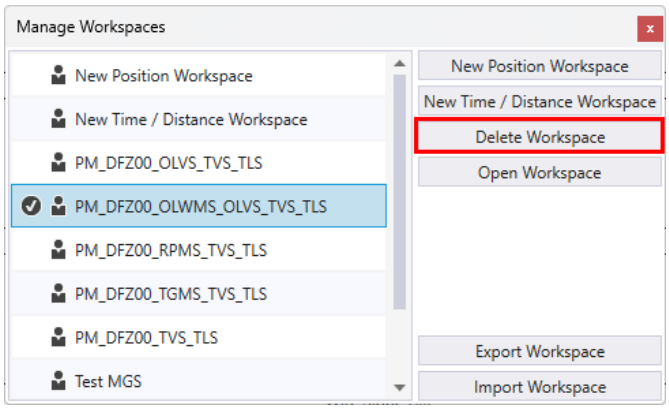
2.3.2. Rename workspace

Workspaces can be renamed by double-clicking on the respective name. This opens a text box in which the name can be changed.



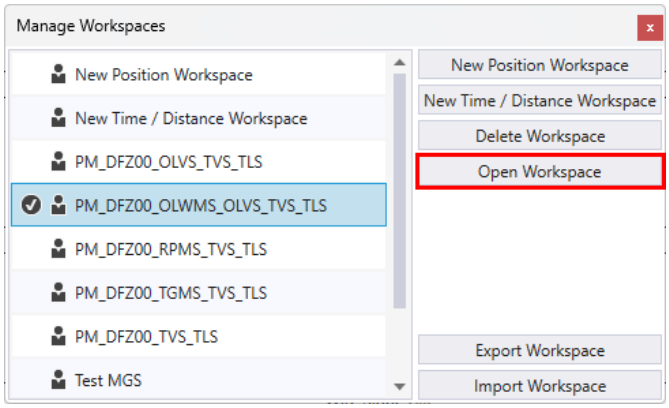
2.3.3. Delete workspace

A workspace can be deleted using the ‘Delete Workspace’ function. Deletion is permanent and cannot be undone.



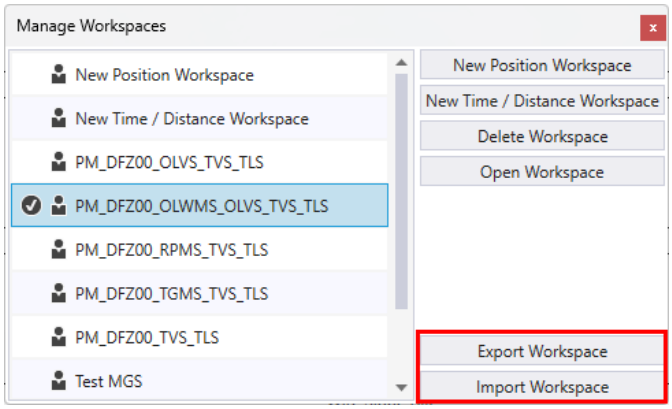
2.3.4. Open workspace

With the 'Open Workspace' function, the desired workspace can be loaded directly from this screen. The 'Manage Workspaces' dialog is closed and the new workspace is displayed.



2.3.5. Export/import workspace

The 'Export Workspace' function can be used to export the desired workspace to a file (.vws). All views contained in the workspace are saved and can be re-imported with the 'Import Workspace' function. This functionality is ideal for creating a backup of a workspace or sharing a specific workspace with other RCM-DX Viewer users.

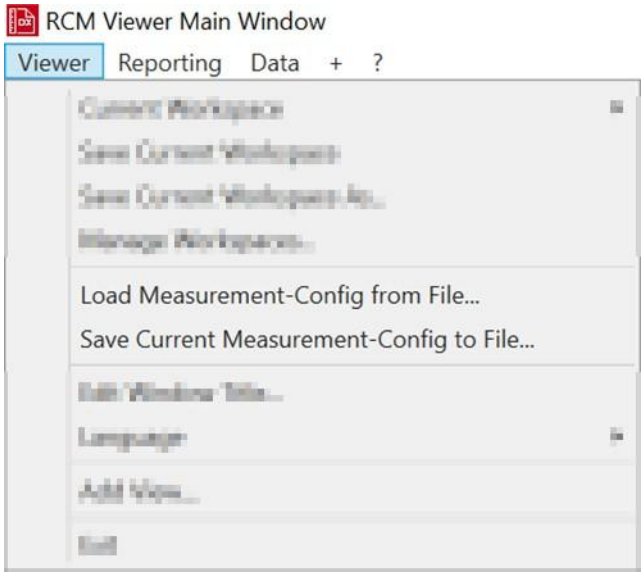


3. Measurement configuration

In a measurement configuration, the ‘Save Current Measurement-Config to File...’ function saves the current workspace together with all data-dependent settings. These include:

- the loaded measurement runs or files
- the navigation settings (current position and zoom level)
- the settings of the 'Loaded Data’ view:
    - Order of sections (master section, comparison sections)
    - Set Colours
    - Set offsets

Measurement configurations can be saved in files with the extension .vmc and reloaded from them with the function ‘Load Measurement-Config from File...’.



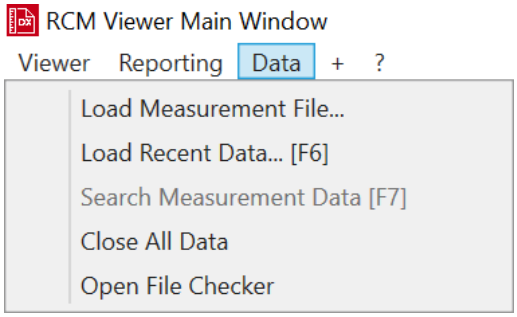
Measurement configurations can be used, for example, to send the state of an analysis to another person. When opening such a file, the corresponding data is loaded (as long as access to the corresponding directories is guaranteed) and the RCM- DX Viewer is rebuilt as it was saved by the creator. The user can see 1:1 what the creator saw when saving it on the screen.

Note: If the .rcmdx file extension is associated with the default program (RCM-DX Viewer) in the operating system, double-clicking the file automatically opens the RCM-DX Viewer, loads all the information and displays it as saved.

4. Data

What is data in the RCM-DX Viewer? Data refers to the information stored in the RCM-DX file format. This information is collected by diagnostic vehicles and stored here for the visualisation with the RCM-DX Viewer.

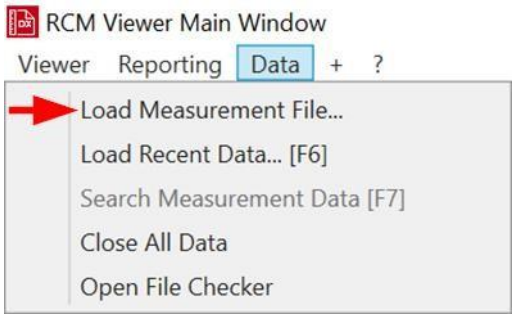
This section describes the functions related to data.



Please note: The Search Measurement Data [F7] function is only available in the online version of the RCM Viewer.

4.1. Load measurement file

With the 'Load Measurement File...' function, measurement files (.rcmdx) can be loaded and opened in the RCM-DX Viewer.

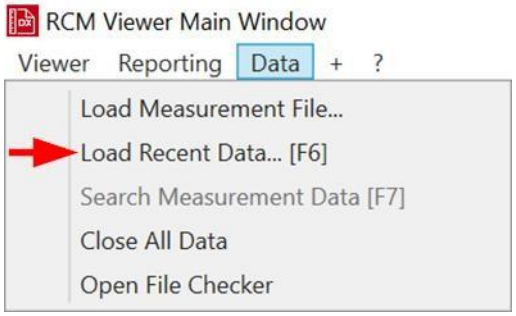


After pressing the menu item, the standard Open-File dialog is opened by the operating system, in which the desired file can be searched for and opened.

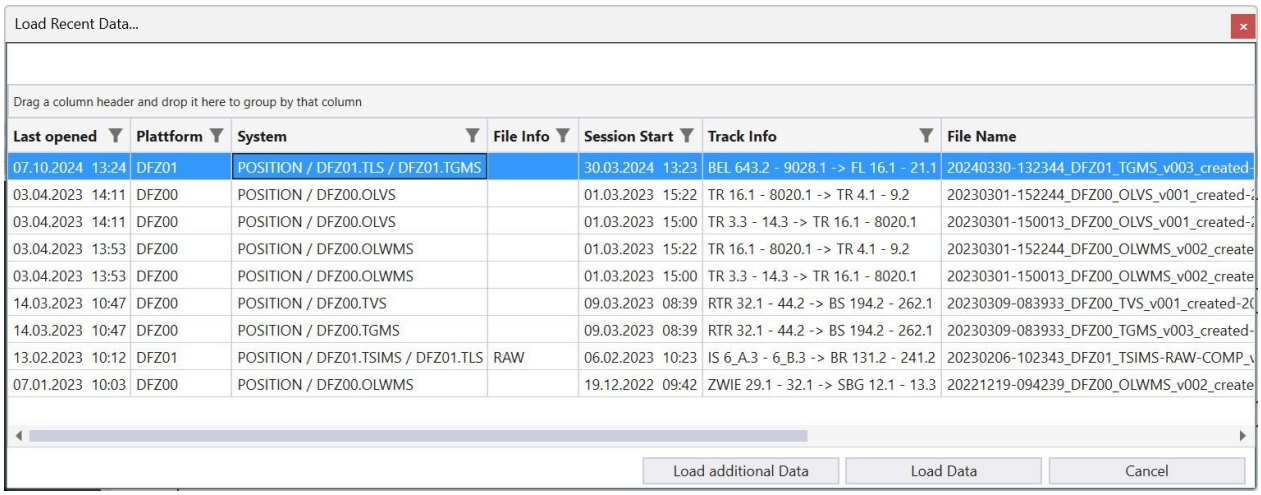
Note: You can open several .rcmdx files at once.

4.2. Load recent measurement files

If you want to reload files that have recently been opened with the RCM-DX Viewer, you can do so using the 'Load Recent Data... [F6]' function.



By clicking on the menu item or pressing the 'F6' key, a dialog is displayed listing the files that were last opened by the user with the RCM-DX Viewer.

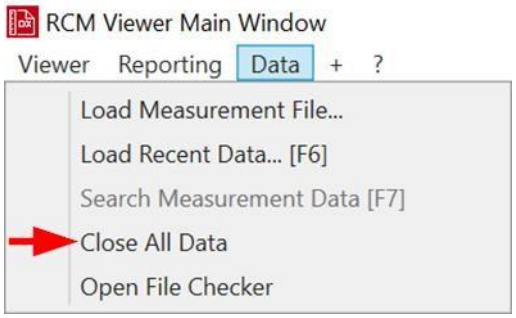


Last opened	Plattform	System	File Info	Session Start	Track Info	File Name
07.10.2024 13:24	DFZ01	POSITION / DFZ01.TLS / DFZ01.TGMS		30.03.2024 13:23	BEL 643.2 - 9028.1 -> FL 16.1 - 21.1	20240330-132344_DFZ01_TGMS_v003_created-
03.04.2023 14:11	DFZ00	POSITION / DFZ00.OLVS		01.03.2023 15:22	TR 16.1 - 8020.1 -> TR 4.1 - 9.2	20230301-152244_DFZ00_OLVS_v001_created-2
03.04.2023 14:11	DFZ00	POSITION / DFZ00.OLVS		01.03.2023 15:00	TR 3.3 - 14.3 -> TR 16.1 - 8020.1	20230301-150013_DFZ00_OLVS_v001_created-2
03.04.2023 13:53	DFZ00	POSITION / DFZ00.OLWMS		01.03.2023 15:22	TR 16.1 - 8020.1 -> TR 4.1 - 9.2	20230301-152244_DFZ00_OLWMS_v002_create
03.04.2023 13:53	DFZ00	POSITION / DFZ00.OLWMS		01.03.2023 15:00	TR 3.3 - 14.3 -> TR 16.1 - 8020.1	20230301-150013_DFZ00_OLWMS_v002_create
14.03.2023 10:47	DFZ00	POSITION / DFZ00.TVS		09.03.2023 08:39	RTR 32.1 - 44.2 -> BS 194.2 - 262.1	20230309-083933_DFZ00_TVS_v001_created-20
14.03.2023 10:47	DFZ00	POSITION / DFZ00.TGMS		09.03.2023 08:39	RTR 32.1 - 44.2 -> BS 194.2 - 262.1	20230309-083933_DFZ00_TGMS_v003_created-
13.02.2023 10:12	DFZ01	POSITION / DFZ01.TSIMS / DFZ01.TLS	RAW	06.02.2023 10:23	IS 6_A.3 - 6_B.3 -> BR 131.2 - 241.2	20230206-102343_DFZ01_TSIMS-RAW-COMP_v
07.01.2023 10:03	DFZ00	POSITION / DFZ00.OLWMS		19.12.2022 09:42	ZWIE 29.1 - 32.1 -> SBG 12.1 - 13.3	20221219-094239_DFZ00_OLWMS_v002_create

Any number of files can be selected and loaded here. The ‘Load Data’ button closes files that have already been opened and only the select ones are loaded into the RCM-DX Viewer.
If you want to add more files to the files that have already been opened, click on the ‘Load additional data’ button.

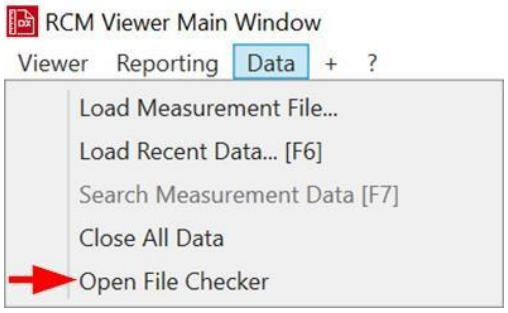
4.3. Close all files

The ‘Close All Data’ function closes all files opened in the RCM-DX Viewer and clears all data in the views (see Chapter 6 Views).

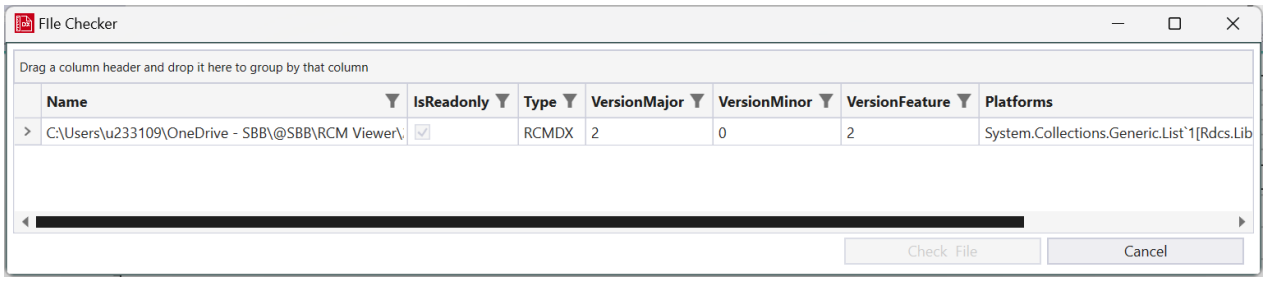


4.4. Open File Checker

With the File Checker (Open File Checker) the files opened in the RCM-DX Viewer can be checked for their validity and possible errors in the event of problems.

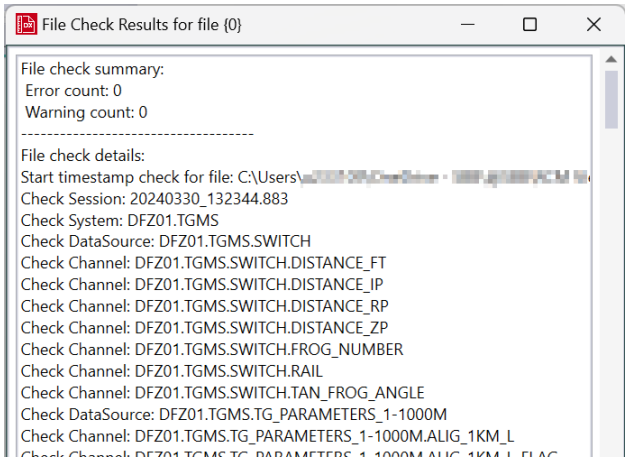


One measurement file opened in the viewer is displayed per line.



If a measurement file is selected, it can be checked by pressing the ‘Check File’ button.

The result is an excerpt with any errors or warnings from the inspected file. No problems were found in the example below.

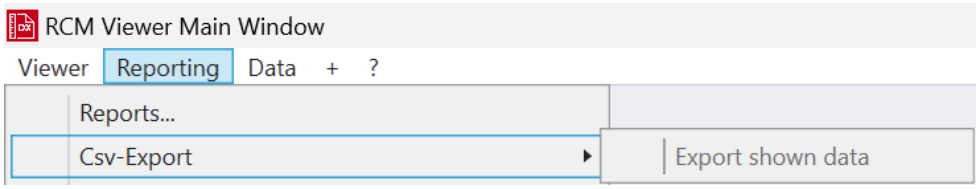


5. Reports (reporting)

The RCM-DX Viewer has two types of reports. The first is “Reports” and the second is “CSV-Export”. The Reports are not described here, as they are not used in the RCM-DX Viewer. The operation of the CSV-Export is described in more detail in the following section.

5.1. CSV-Export

The RCM-DX Viewer offers the option to export displayed measurement data as CSV files.



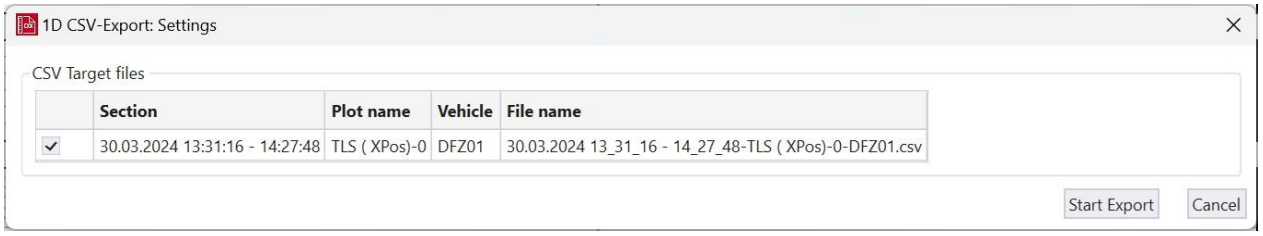
Data displayed in diagrams is exported in the 'Values over time / position' view. A separate CSV file is created for each of these plot views. Each channel in this view, whether in several diagrams or combined in one, is exported to the CSV file as a separate column.

The 'Export Shown data' menu item exports data in the displayed section of the navigation. 'Displayed' therefore refers to the area of the navigation bar, not to which data is displayed in the plots – exported will only be data and channels that are displayed in the diagrams in the workspace. Similarly, only data from the current reference run is exported (if several sessions or sections are loaded).

The values are exported in the unit set for the respective plot in the 'Values over time / position' view.
If no diagrams / plots are configured in the workspace, or only those for which no data can be displayed with the loaded measurement runs, no export can be made and a corresponding error message appears.
The export dialog depends on the navigation mode in which the viewer is currently in (time/distance mode or position mode) – in time/distance mode, an export frequency can be selected by the user.

5.1.1. Export in position mode

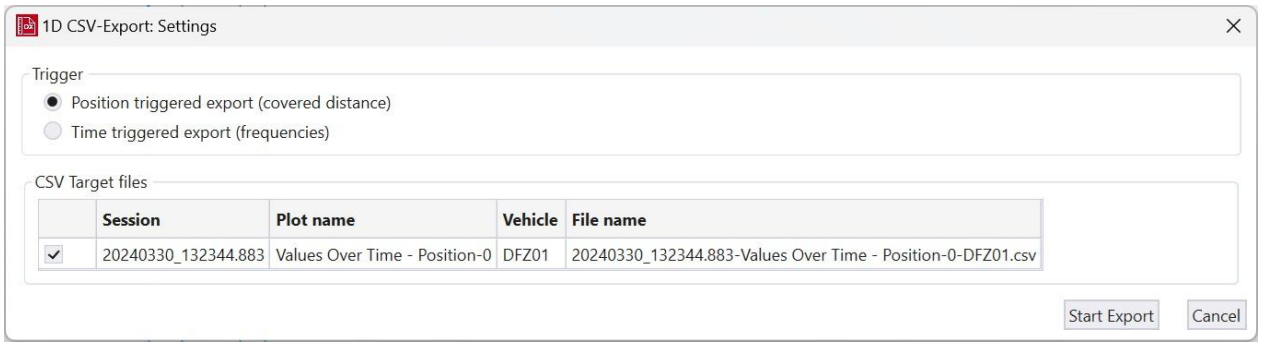
For each combination of section view and plot view, a filename according to the schema 'Sectionname-Plotname-ExportNumber-Vehiclename.csv' is suggested in the table. The user can edit this in the last column of the table. If a file with the same name already exists during the export, a sequence number is appended in brackets, e.g. DateiName (1).csv.
The checkboxes in the first column allow the user to select which CSV files to export. All files are selected here by default.
In position mode, the positioning system specifies the frequency at which the data is exported.



5.1.2. Export in Time/Distance Mode

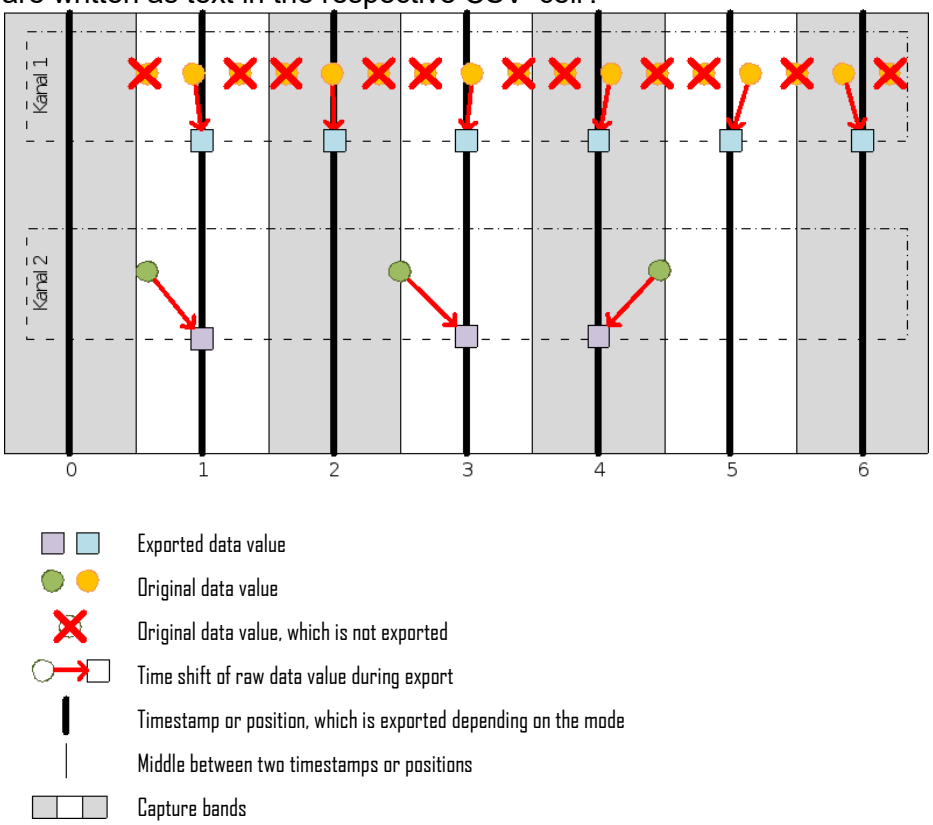
In time/distance mode, the user also has the option to select the export frequency, in addition for it to be specified by the positioning system. The user can then change the data frequency for each plot view in the table with the plot views. The selected frequency is used for the export for all open sessions for the respective plot view.

For each combination of session and plot view, a filename according to the schema 'Sessionname-Plotname-ExportNumber-Vehiclename-Frequency.csv' in the table. The user can edit this in the last column of the table. The frequency is only used if the export trigger is set to 'Time-triggered Export'.



5.1.3. Alignment of the exported data values

For each point in time or position to be exported, the best-fitting data value of a channel is used. For this purpose, a capture window is defined at each point in time or position. For a position, the capture window is equal to half the distance from the neighbouring positions around the current position. In time mode, the capture window is equal to half the period before and after the current point in time. A capture window encompasses the lower limit and excludes the upper limit. Exported channel data / measured values are therefore never interpolated.
Of all data values in a capture window, the one closest to the actual time or position is selected. If two data values are the same distance from the time or position, the data value measured earlier is used.
If there is no measurement value within the capture window, no value is exported. This means that the corresponding CSV ‘cell’ is left blank.
The data values are exported without conversion. Special values such as NaN, INF or similar are written as text in the respective CSV ‘cell’.



5.1.4. Interpolation of positions

In contrast to the measurement values, which are never interpolated (as explained and visualised in Section 2.11.4), for the positions with frequency-triggered export, the

positions to interpolate. Components of the position are interpolated linearly. This includes the timestamp, the covered distance and the track offset. The line offset is not interpolated.

5.1.5. Structure of the exported CSV file

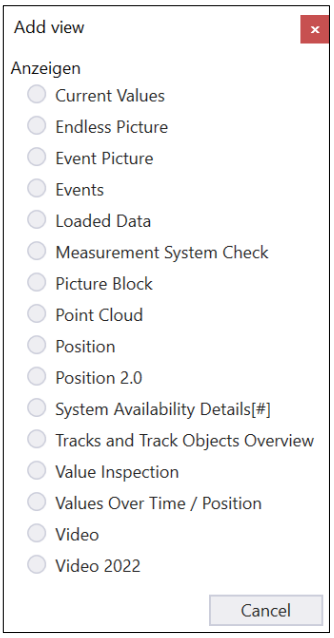
```
#Output Dateiname;10.12.2019 07_52_58 - 08_49_26-TGMS-1-DFZ00.csv
#Input Dateiname(n);\\mudnas\rdpf_int_internal\REPORTS\2019\12\DFZ\TGMS\20191210_075251_TGMS_created-20191212_112947.h5,\\mudnas\rdpf_int_internal\REPORTS\2019\12\DFZ\RPMS\20
#Session Name;20191210_075251_13.ok
#Export Zeitpunkt;11.06.2020 08:17:21
#Exporter;RCM DX (Offline) Viewer
#Exporter Version;1.66.0.0
#DataExport Library Version;0.5.1.0

#Time-UTC;Time [ns];Covered distance [m];Line name;Line offset [km];Track name;Track offset [m];DFZ00.TGMS.TGPAR_111.GAUGE [mm];DFZ00.TGMS.TGPAR_111.CANT [mm]
10.12.2019 07:30:08;1575963008516730000;34318;Löchligut - Wanzwil - Rothrist West;31.571720123291;AESP 284.2 - WANZ 112.2;19212.33203125;1.23;-0.96
10.12.2019 07:30:08;1575963008525160000;34318.25;Löchligut - Wanzwil - Rothrist West;31.5719699859619;AESP 284.2 - WANZ 112.2;19212.58203125;1.33;-0.94
10.12.2019 07:30:08;1575963008533644000;34318.5;Löchligut - Wanzwil - Rothrist West;31.5722198486328;AESP 284.2 - WANZ 112.2;19212.83203125;1.39;-0.78
10.12.2019 07:30:08;1575963008542116000;34318.75;Löchligut - Wanzwil - Rothrist West;31.5724697113037;AESP 284.2 - WANZ 112.2;19213.08203125;1.43;-0.67
10.12.2019 07:30:08;1575963008550601000;34319;Löchligut - Wanzwil - Rothrist West;31.5727195739746;AESP 284.2 - WANZ 112.2;19213.33203125;1.39;-0.53
10.12.2019 07:30:08;1575963008559088000;34319.25;Löchligut - Wanzwil - Rothrist West;31.5729675292969;AESP 284.2 - WANZ 112.2;19213.580078125;1.38;-0.26
10.12.2019 07:30:08;1575963008567549000;34319.5;Löchligut - Wanzwil - Rothrist West;31.5732173919678;AESP 284.2 - WANZ 112.2;19213.830078125;0.9899999;-0.13
10.12.2019 07:30:08;1575963008576034000;34319.75;Löchligut - Wanzwil - Rothrist West;31.5734672546387;AESP 284.2 - WANZ 112.2;19214.080078125;1.35;-0.06
10.12.2019 07:30:08;1575963008584529000;34320;Löchligut - Wanzwil - Rothrist West;31.5737171173096;AESP 284.2 - WANZ 112.2;19214.330078125;1.44;-0.05
10.12.2019 07:30:08;1575963008593008000;34320.25;Löchligut - Wanzwil - Rothrist West;31.5739669799805;AESP 284.2 - WANZ 112.2;19214.580078125;1.43;-0.06
10.12.2019 07:30:08;1575963008601435000;34320.5;Löchligut - Wanzwil - Rothrist West;31.5742168426514;AESP 284.2 - WANZ 112.2;19214.830078125;1.43;-0.11
10.12.2019 07:30:08;1575963008609928000;34320.75;Löchligut - Wanzwil - Rothrist West;31.5744667053223;AESP 284.2 - WANZ 112.2;19215.078125;1.46;-0.18
```

An exported CSV file has a 'header' section with metadata. Then, after a blank line, the names and units of the columns follow. Each row of data corresponds to a position or a time. The first columns always contain the default data, while an additional column is added for each channel in a plot view. Empty data values are not marked separately, but the 'cell' is simply left blank.

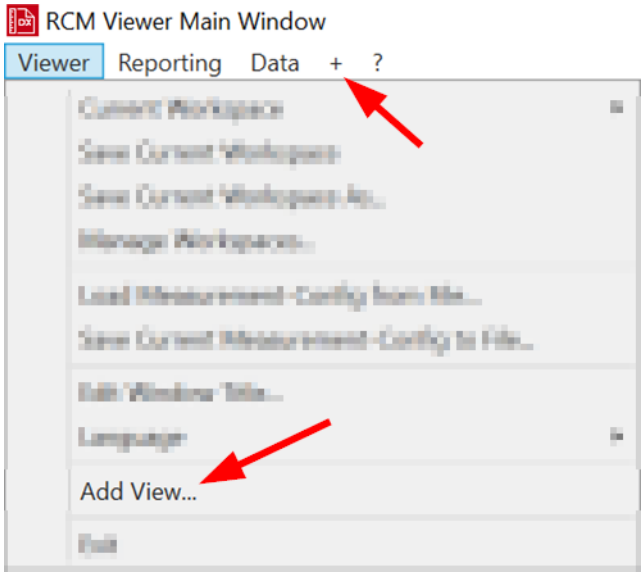
6. Views

A view is used to display information. Information can be e.g. values, pictures, positions, lists or videos. Each view has a specific purpose to display the information to the user. Views can be arranged on the screen in so-called workspaces. See Chapter 2 (Workspaces). The size and position of each view can be selected individually. The same view can be contained several times in the same workspace, e.g. several pictures. To add the same view again, the corresponding view can be selected again during the selection.

Here is a selection of the possible views in the RCM-DX Viewer:



The selection of the available views can be found in the menu line under the menu item '+' (plus sign) or under the menu item 'Viewer' with the function 'Add View...'.



The nature and function of the individual views are described in detail in the following sections.

6.1. View: Current Value

The 'Current Value' view displays values of one-dimensional variables. The designation and unit for each variable are displayed.

Reference to the navigation control: The view shows the values at the current position of the navigation control.

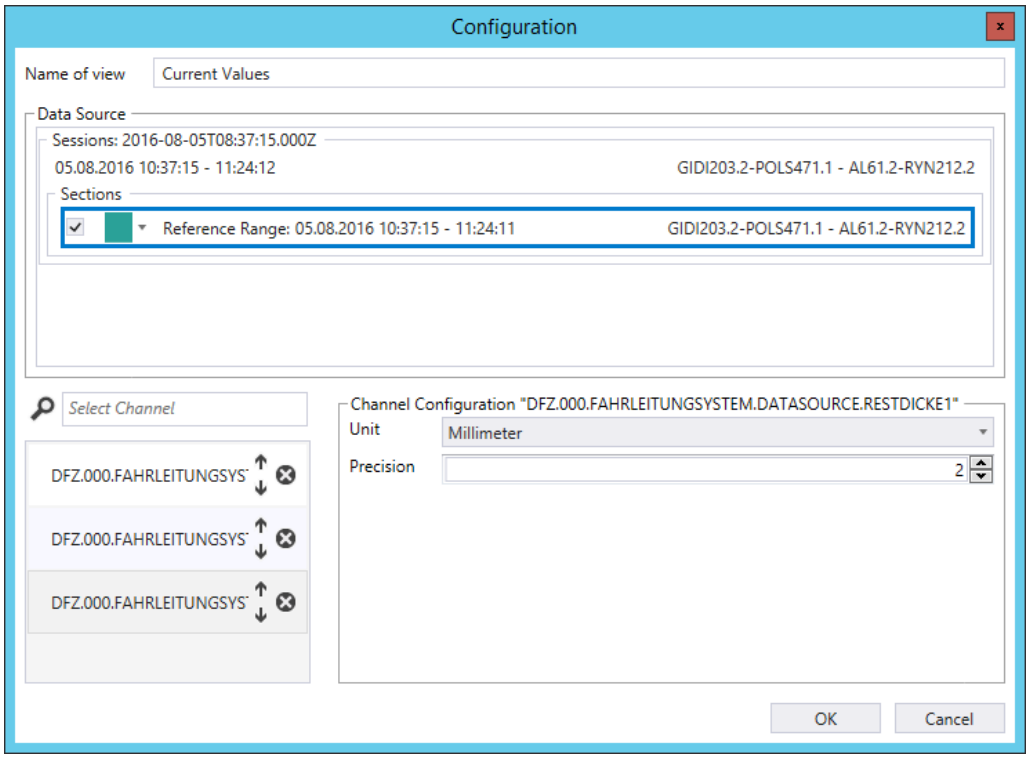
If the value of a parameter is outside the valid range (if defined), the limit value violation is represented by a different background colour.

Difference between time/distance mode and position mode:

- In time/distance mode, a list of channels can be defined in the configuration. The view then shows the current value for each channel.
- In position mode, both a list of channels and a selection of measurement runs can be defined in the configuration. The view then displays the values of all selected measurement runs for each channel. This means that the values of the different measurement runs can be compared per channel.

6.1.1. Configuration (current value)

The following figure shows the configuration dialog of the view in position mode:

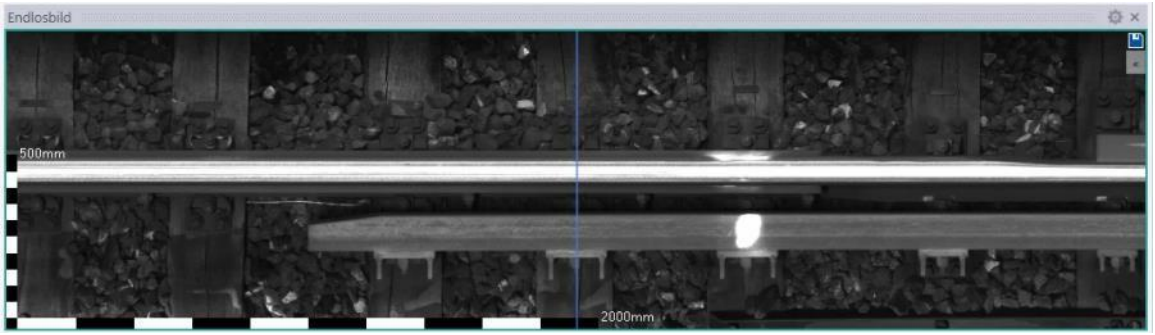


The following can be set for each channel:

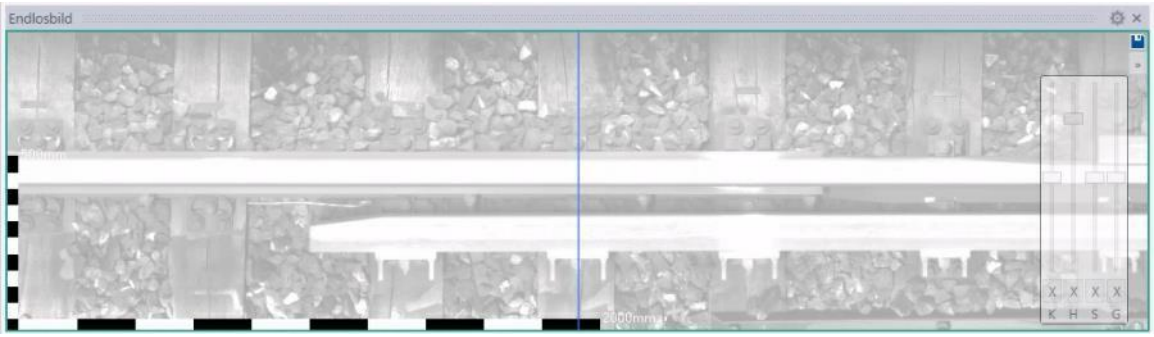
- The desired unit
- The number of decimal places displayed

6.2. View: Endless Picture

This view represents the same metrics as the view image and combines the images, if they belong together in time, to form a long (theoretically ‘endless’) picture.



Reference to the navigation control: The view shows the image at the current position of the navigation control.
The current position is visible as a blue bar in the image. The blue bar cannot be moved in this view. The displayed image section can be determined using the mouse. The save icon ( ) saves the currently displayed image section.
The image editing functions can be displayed via the edit icon ( )
These allow you to manually adjust contrast, brightness, saturation and gamma.

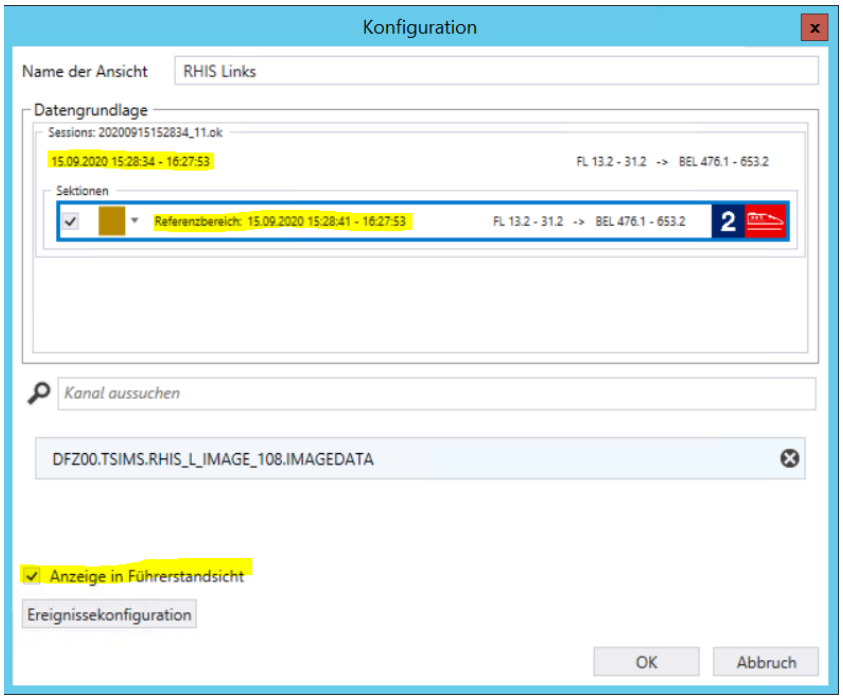


The continuous image view only allows the display of a single image channel.
Additional position mode:

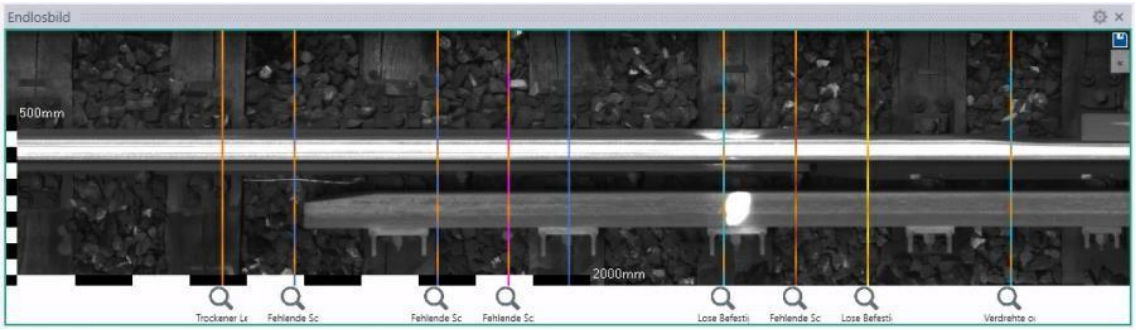
- In position mode, a measurement run has to be selected next to the measurement channel for which data should be displayed. The view then receives a frame in the colour of the measurement run as defined in the ‘Loaded Data’ view.

6.2.1. Configuration (continuous image)

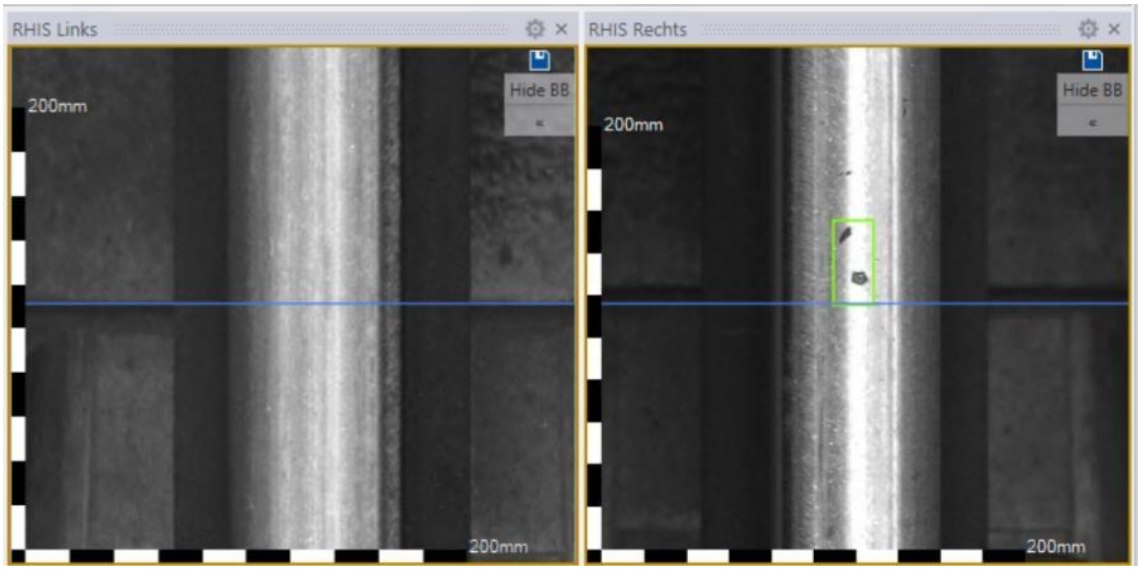
The following figure shows the configuration dialog in a workspace in position mode:



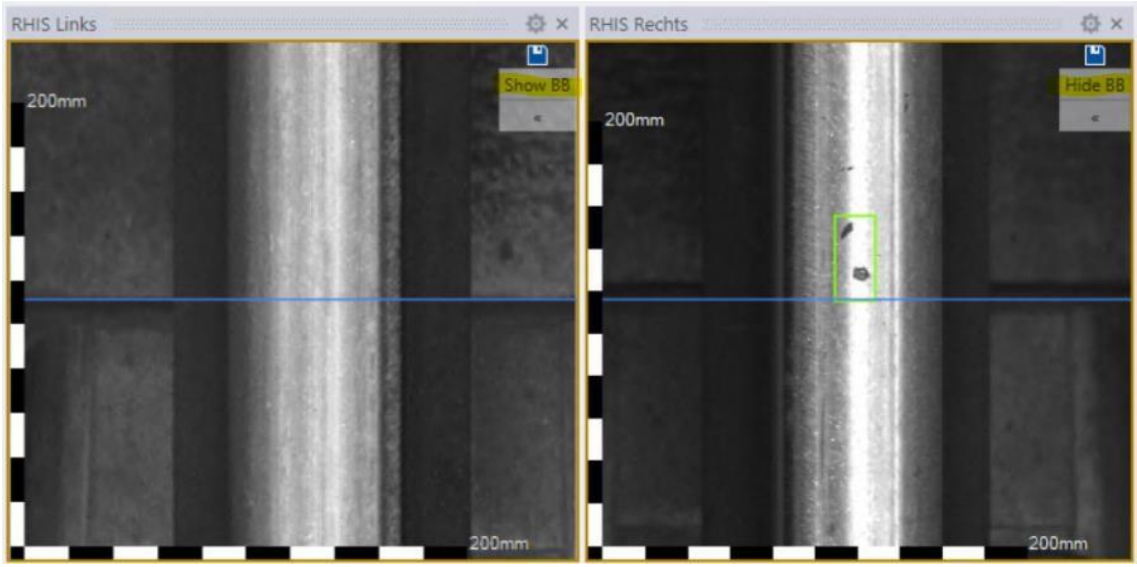
The button ‘Event Configuration’ allows you to set the display of events in the continuous image view. The functions here are identical to those in the navigation control.
The continuous view - with events activated - looks like this:



Driver’s cab view: the option ‘Display in driver’s cab view’ shows the continuous images running from bottom to top:



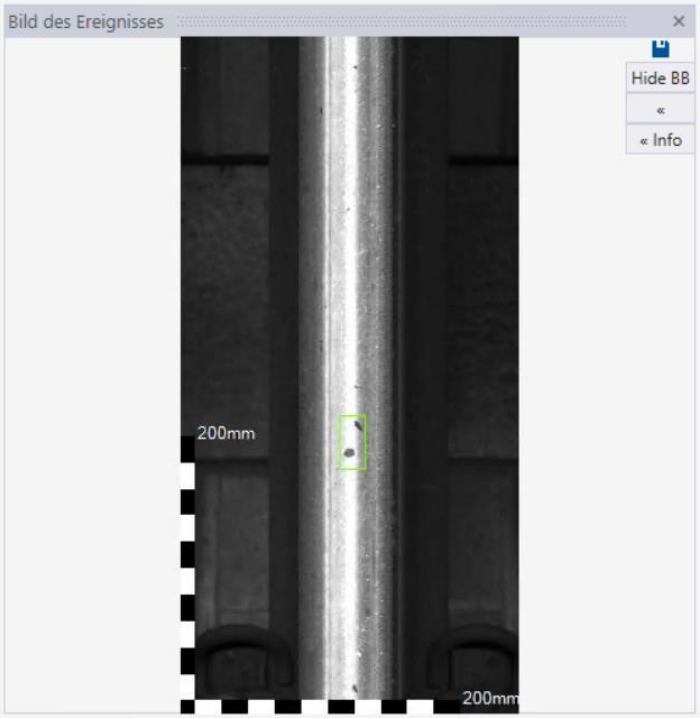
Displaying/hiding the bounding boxes: by clicking on the ‘Show BB’ / ‘Hide BB’ button, the bounding box (a green frame around the damage site) can be displayed or hidden.



Scale
In order to determine the size of defects and objects in pictures, scales have been applied in vertical and horizontal directions to the edges of the pictures. These change accordingly in the zoom. The number next to and above the scale indicates the length of the entire black and white bar. A black or white element of the scale is as long as 1/10 of the total length of the scale.

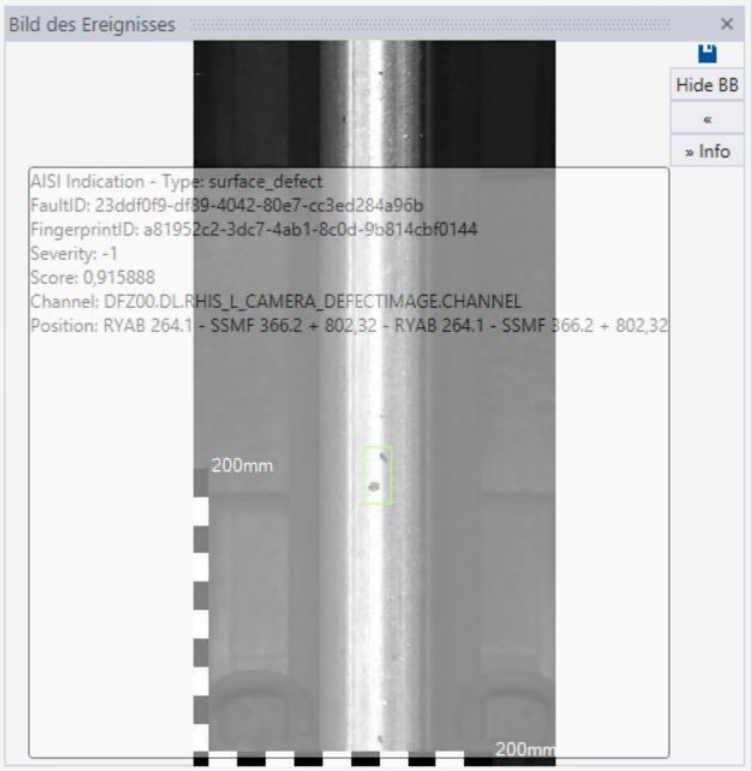
6.3. View: Event Picture

This view shows an image of an event. The event can be selected in the event list 2.10.1.



The image remains visible until a new event is selected. If there is no image for this event, the information ‘No image’ is displayed and the view remains grey.
The displayed image section can be determined using the mouse. The save icon (💾) saves the currently displayed image section.
You can use ‘Show BB’ / ‘Hide BB’ to show and hide the bounding box (a green frame around the damage site) as with the continuous images.

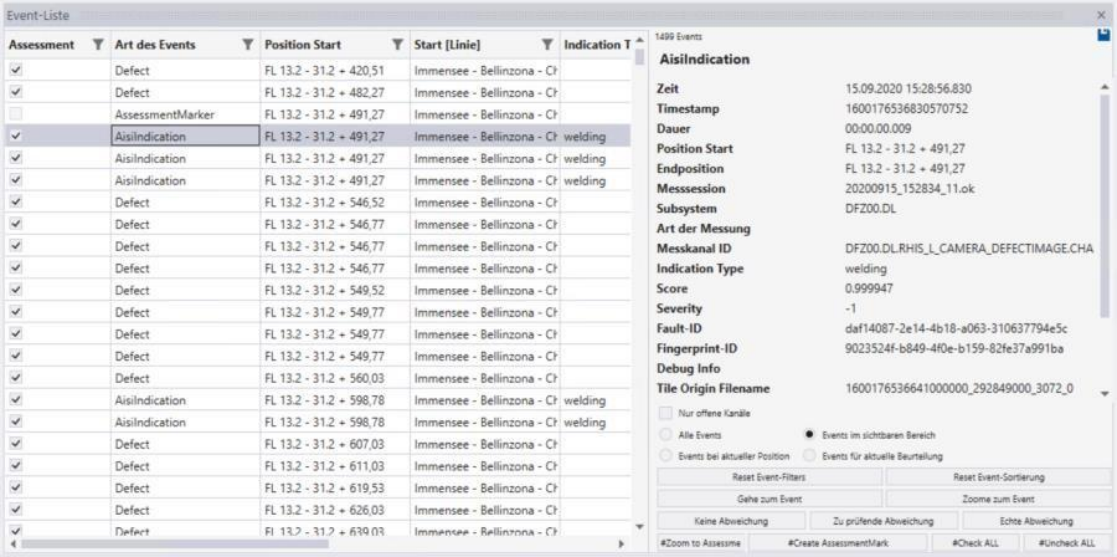
In addition, you can display information about the image in this view:



The information shows information about the event, from which channel the image was taken and the position information (start and end).

6.4. View: Events

The Events view displays a table of events. Details of the selected event are displayed in the right pane.

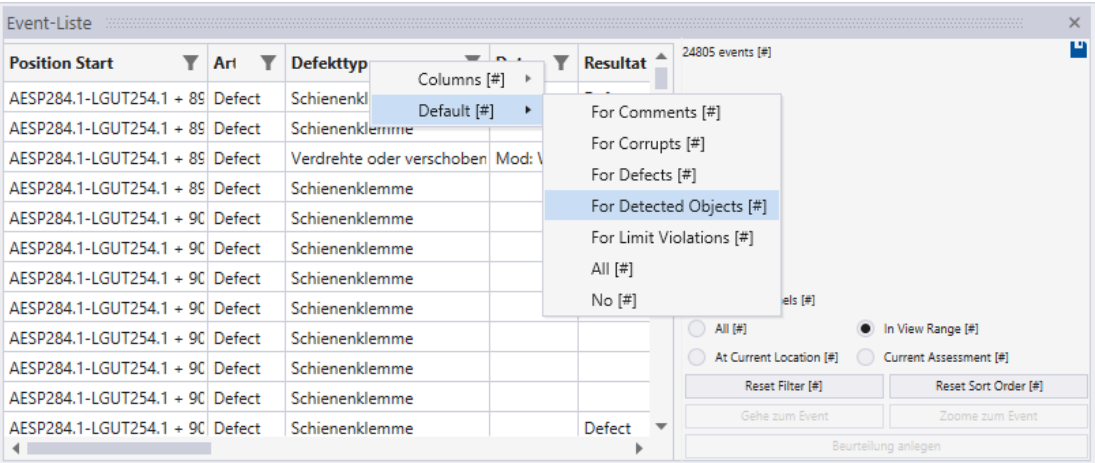


Related to the navigation control: The view always shows all events, regardless of the area selected in the navigation control.

Difference between time/distance mode and position mode: In time/distance mode, the events of the reference measurement run are displayed. In position mode, the events of all scheduled measurement runs are displayed.

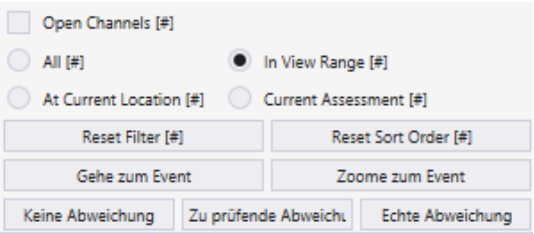
The view offers the following functions:

- The table can be filtered (click on filter icon in column header) and sorted (click on column header)
- The displayed columns can be selected (right-click on column header). It is possible to show/hide all columns individually, or default settings are available.
- The list of events can be exported as a CSV file (symbol 💾)



The selected columns as well as sorting and filtering are saved in the workspace.

Filter functions
Below the event details, various filter functions are available:



These filter functions are:

- ‘Open Channels’: Show only events related to a displayed channel.
- ‘In View Range’: Show only events that are in the visible range of the navigation view.
- ‘At Current Location’: Only events that occur at the current position of the blue bar (i.e. the blue bar is between event start and end).
- ‘Current Assessment’: Show only events that relate to the currently selected infrastructure assessment.

The other buttons have the following functions:

- ‘Reset Filter’: Resets all filters in the column headers.
- Reset Sort Order: Resets the sorting to Default. Default means that the events are displayed in the order of the start position.
- ‘Go to Event’: Sets the position of the blue bar to the start of the selected event (even if an event is newly selected).
- ‘Zoom to event’: Sets the visible area in the navigation view to duration of events (also happens when double-clicking the event).
- ‘Create assessment with deviation’: Allows the creation of a new infrastructure assessment related to the selected event.

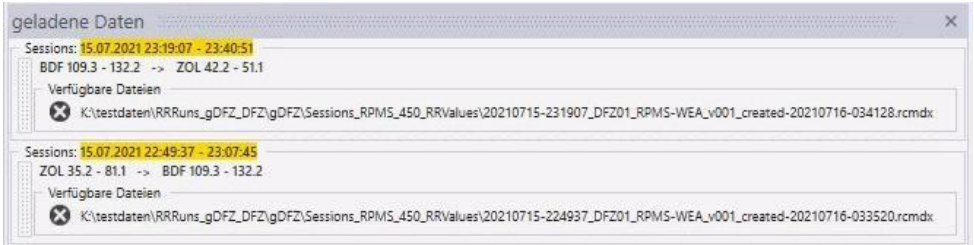
Relationship Event and Assessment
An assessment can have a relationship to none or to several events. All these events are marked in red once the assessment has been selected.

Note: It is possible that events marked in red are outside the events currently displayed (for many events not all can be visible in the list and are outside the scroll area). The filter function ‘Current Assessment’ described above can be helpful here.

6.5. View: Loaded Data

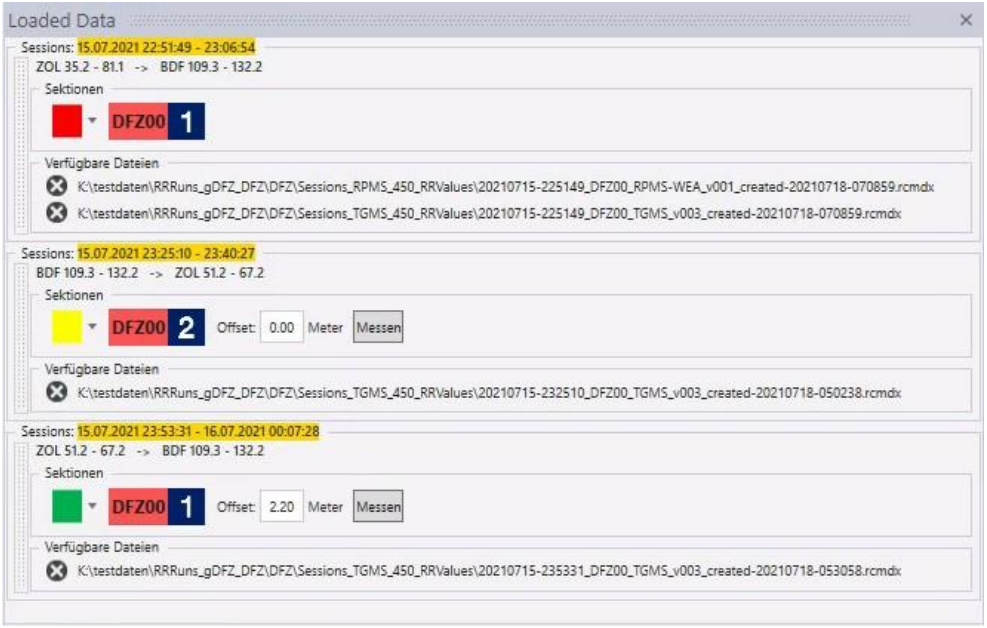
This view shows the currently loaded data.

Time/distance mode
If a workspace is open in time/distance mode, the Loaded Data view shows all available measurement runs.



The measurement runs can be sorted using drag and drop. To do this, they are grabbed with the mouse on the left edge and dragged to another list position. The topmost measurement run in the list becomes the reference measurement run. The area of the reference measurement run to be displayed can then be selected via the navigation control.
Loaded measurement data can be closed again by clicking on the icon (⊗).

Position mode
If a workspace with position mode is open, the ‘Loaded Data’ view shows all available measurement runs. However, each measurement run can be divided into one or more measurement run sections (‘sections’). This is the case if the same route has been crossed several times during a measurement run. Each crossing leads to a new measurement run section. An icon for each measurement run section indicates how the vehicle’s orientation was during this measurement. The number in the icon indicates which driver’s cab (1 or 2) was at the front during the journey.



The measurement runs can be sorted using drag and drop. To do this, they are grabbed with the mouse on the left edge and dragged to another list position. The topmost measurement run in the list becomes the reference measurement run. The area of the reference measurement run to be displayed can then be selected via the navigation control.

The following settings can also be made:

- In addition to the reference measurement run, the list item also shows the order of the ancillary runs.
- Each measurement run section can be assigned a colour. The views show the data of the corresponding section in this colour.
- For ancillary journeys, a positioning offset relative to the reference journey can be specified; this offset is applied to all data for the ancillary journey.
- The offset can be measured directly in a ‘Values over Time / Position’ view.
- Loaded measurement data can be closed again by clicking on the icon (⊗)

6.6. View: System test (Measurement System Check)

The 'System Test’ view is available in the RCM-DX Viewer. In order for a system test to be carried out, you need access to the Analytics Service. You select the environment in which to do this from the start on the start screen. There are basically three environments to choose from here: PROD, INT and DEV.

As a user, you will primarily work on INT or PROD. The DEV environment is, as the name suggests, meant for the developers.
Then you either create a new system test or load an existing one from an XML file.

6.6.1. Concepts

The implementation of the system test introduced a number of new concepts from the viewer's point of view, which are explained here.

Logical and Physical Measurement System

We distinguish between a 'logical' measurement system which is a higher-level measurement system, e.g. 'TGMS' for Track Geometry Measurement System. A logical measurement system comprises one or more 'physical' measurement systems. The physical measurement systems are those which effectively appear in RCM-DX files as 'System' (e.g. DFZ00.TGMS). Since the system test has an abstract view of the systems and is independent of the platform/vehicle, this concept of the logical measurement system was introduced.

Measurement parameter

The system test (and also the measurement data comparison) uses the concept of a 'measurement variable'. This, in turn, is a view that is independent of the vehicle and the specific measurement channel, or combines it. The corresponding measurement channels of the different vehicles are then assigned to a measurement variable. Currently, the left/right variants of measurement variables are each separate measurement variables. There are therefore, for example, the measurement variables 'direction D0 left' and 'direction D0 right'.

Versioning of DataSources

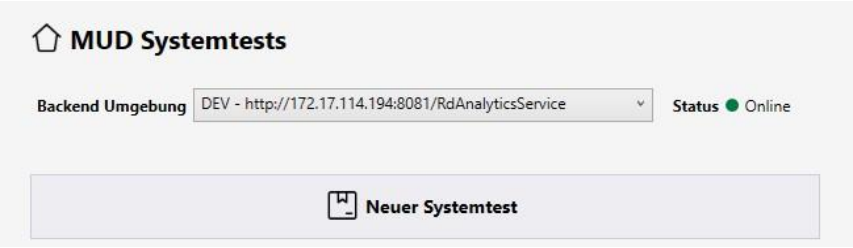
If a DataSource is versioned by its name, such as DFZ00.TGMS.TGPAR_xy, it must be ensured that the parameter configuration is updated when the name of a DataSource (and thus the names of the channels) changes.
This configuration is currently still maintained by the SCS in the Config Service. It is also important to note that this configuration is duplicated: For the system test and the automatic measurement data comparison.

Persistence

A system test is persisted via an external XML file. The 'Save' dialogues in the view create these XML files. These can then be saved, shared and reloaded at a later point in time. An XML system test can either contain only the parameterisation or also the results if the saving has been done from the results view.

6.6.2. Create new system test

If you create a new system test, you are guided through a series of menus to set up the system test initially.

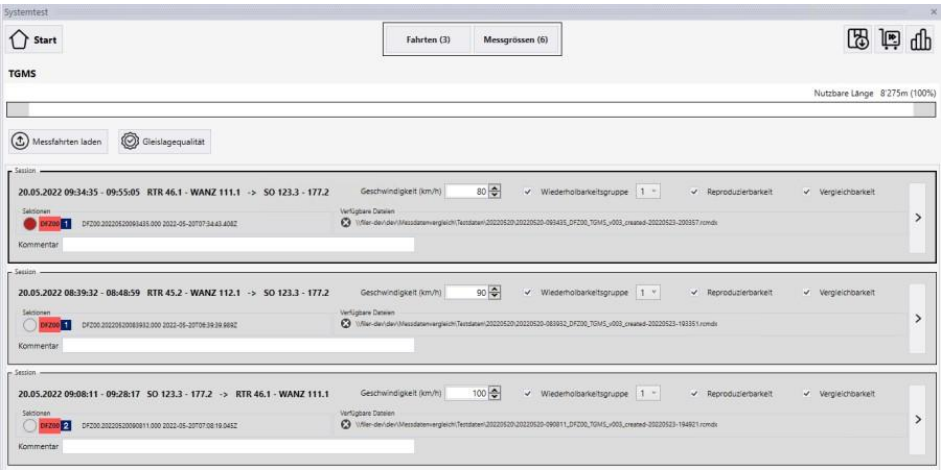


1. The first step is to select the logical measurement system (TGMS, RPMS, etc.)
2. Then there is a file dialog where you can select all RCM-DX files for the system test. If you forget one, you can also upload files afterwards
3. Step 3 is the selection of the parameters. This selection can also be changed afterwards.

6.6.3. Parameterisation

Overview of journeys

In the 'Trips' overview, all trips loaded for the system test are shown. The topmost journey is the reference journey. A journey can be defined as a reference by pressing the round button next to a section.

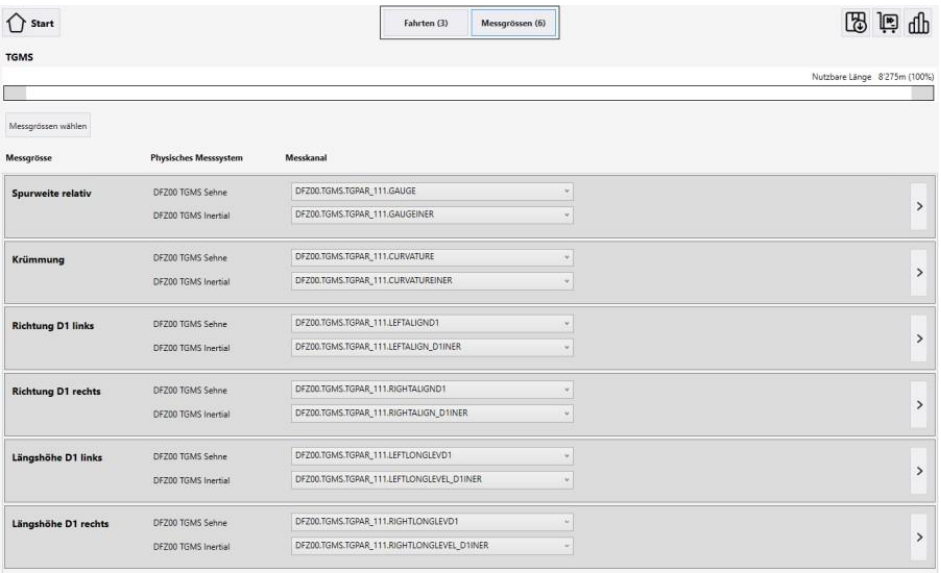


In this view, you define the following parameters for each loaded session and section:

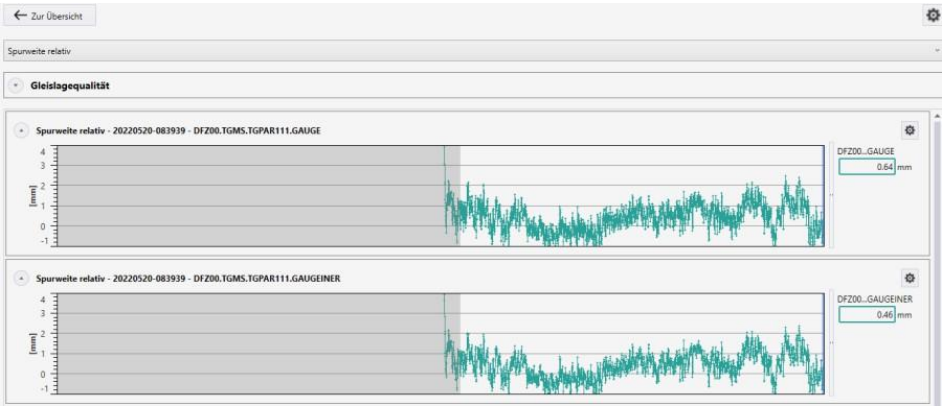
- **Speed** driven: Intended as information for the user only, has no influence on the calculations.
- **Repeatability**: Defines whether repeatability is calculated for this trip. These trips can be grouped using the numbers.
- **Reproducibility**: Enable reproducibility for this trip.
- **Comparability**: Include systems from this journey for comparability.
- **Comment**: Can be used as additional information by users. Has no influence on the calculation.

Additional trips can be loaded via the 'Load measurement trips' button. However, this can also be done via the general menu. A trip or a file can be removed from the context via the ':cross:' button under the 'Available files'.

Overview of the parameters
In the metrics view, you can select the metrics and see which channels are available for the trips you have just loaded.



Detailed view of measurement signals
The '>' button on the far right of each journey or measurement variable can be switched to the detailed view with the measurement signals. Here the measurement signals are displayed grouped by session or measurement variable.

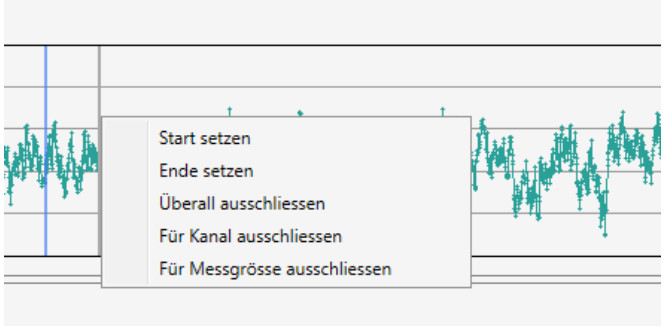


Plot configuration
The plots of the measurement signals can be configured in a similar way to the plots in the well-known 'Values over time/position' view. Important to know here is that the large cogwheel at the top right contains the entire plot configuration, which is also persisted in the workspace. Here you can configure the plots for each group of measurement variables.
In addition, you can also adjust the settings for each individual plot. However, these are not persisted and are lost when you leave the view.

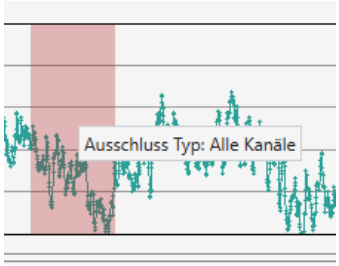
Exclusions
An important feature of this detail view is the ability to define exclusions. Exclusions are areas that should be ignored when calculating the parameters. There are different types of exclusions:

- **Total exclusions** (exclude everywhere) – applied to all trips, metrics and channels.
- **Parameter-specific**: Applies to a specific parameter (i.e. for all channels assigned to this parameter).
- **Channel-specific**: Only applies to a specific channel of a specific measurement run.
- **Start and end exclusions**: Applies like the total exclusions for all measurements and trips, but can only be set once at a time.

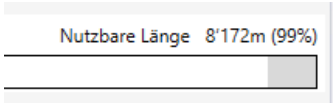
An exclusion can be set via the context menu, by right-clicking on a measurement signal plot.



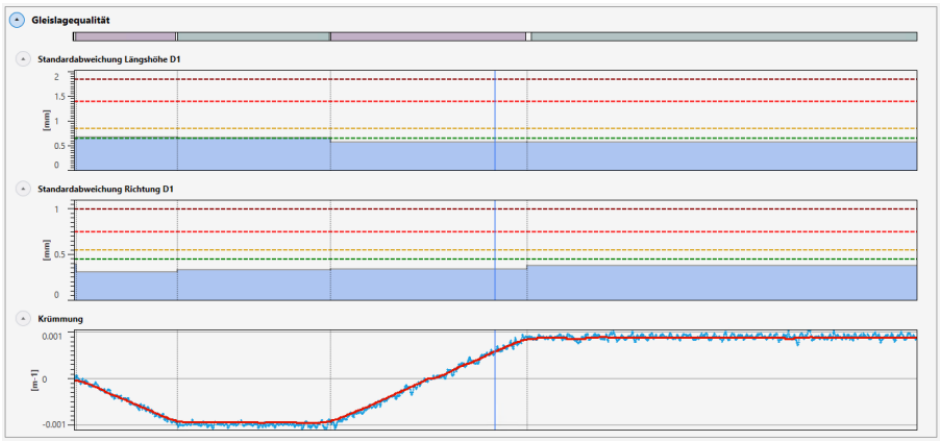
Each type of exclusion has its own colour. The tooltip at mouse-over displays additional details about the exclusion.



The exclusions made are then visible in the plots of the measurement signals. In addition, there is also a view with the exclusions in the overview of the runs and the measurement variables, together with an indication of how much length is available for the system test.

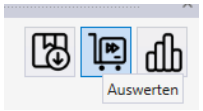


Track alignment quality
The 'Track alignment quality' button in the 'Trips' view calculates the track alignment quality of the current reference run. If the reference run is changed, the results are deleted. The result of the track alignment quality is visible in the detail view of the measurement signals. The limit values for the quality levels are entered in the plot, from bottom (A, green) to top (to E, dark red).



6.6.4. System test Evaluate

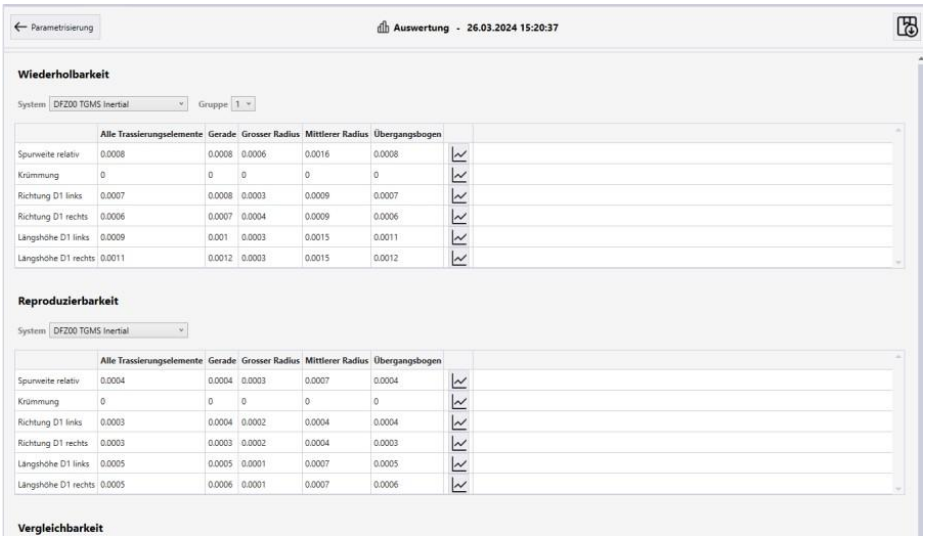
Press the 'Evaluate' button to start the system test. A progress bar appears and finally the message whether the evaluation was successful or not. If successful, the results are then available in the results view.



6.6.5. Results

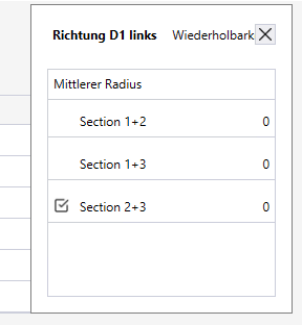
The results of the system test are presented primarily in tabular form, separately for each parameter (repeatability, reproducibility, comparability).

The repeatability is always calculated for pairs of trips. The worst value is then displayed in the table. If you click on a cell in the table, the results for all pairs of trips are visible.



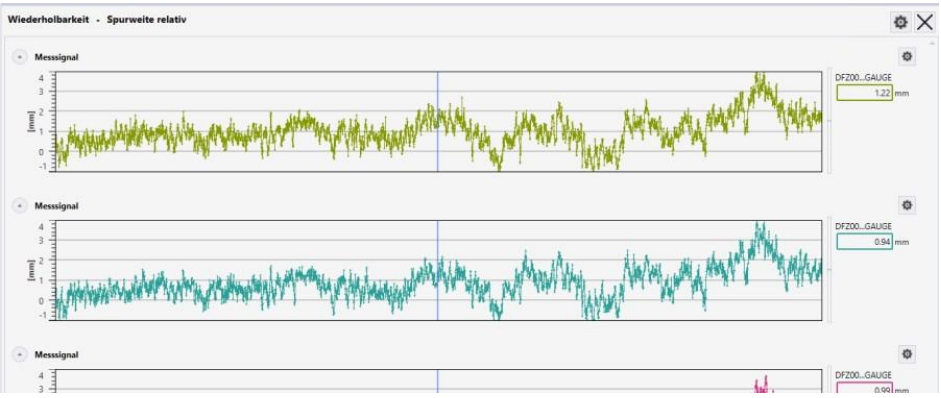
	Alle Trassierungselemente	Gerade	Grosser Radius	Mittlerer Radius	Übergangsbogen
Spurweite relativ	0.0008	0.0008	0.0006	0.0016	0.0008
Krümmung	0	0	0	0	0
Richtung D1 links	0.0007	0.0008	0.0003	0.0009	0.0007
Richtung D1 rechts	0.0006	0.0007	0.0004	0.0009	0.0006
Längshöhe D1 links	0.0009	0.001	0.0003	0.0015	0.0011
Längshöhe D1 rechts	0.0011	0.0012	0.0003	0.0015	0.0012

	Alle Trassierungselemente	Gerade	Grosser Radius	Mittlerer Radius	Übergangsbogen
Spurweite relativ	0.0004	0.0004	0.0003	0.0007	0.0004
Krümmung	0	0	0	0	0
Richtung D1 links	0.0003	0.0004	0.0002	0.0004	0.0004
Richtung D1 rechts	0.0003	0.0003	0.0002	0.0004	0.0003
Längshöhe D1 links	0.0005	0.0005	0.0001	0.0007	0.0005
Längshöhe D1 rechts	0.0005	0.0006	0.0001	0.0007	0.0006



Detailed view
Using the button, you can switch to the detailed view, where results and signals are visible via position (always related to a specific measurement variable).

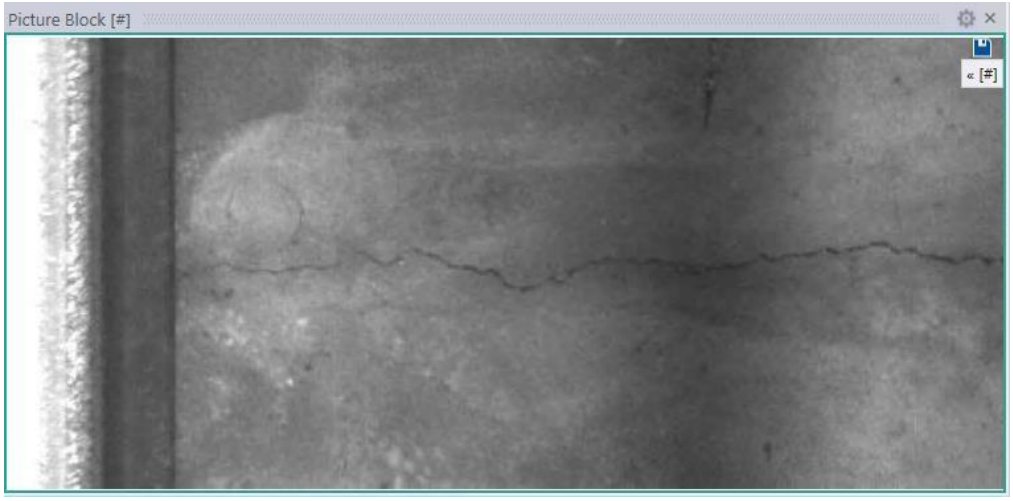
- Repeatability shows the synchronised measurement signals and the difference signals.
- The reproducibility shows the standard deviations in a plot.
- Comparability is shown by the scatter band.



Save
If you save a system test from the results view, the results are also appended. This is different from saving from the parameterisation view, where only the parameterisation is saved.

6.7. View: Image (Picture Block)

This view represents a single measure of the Image Block type.

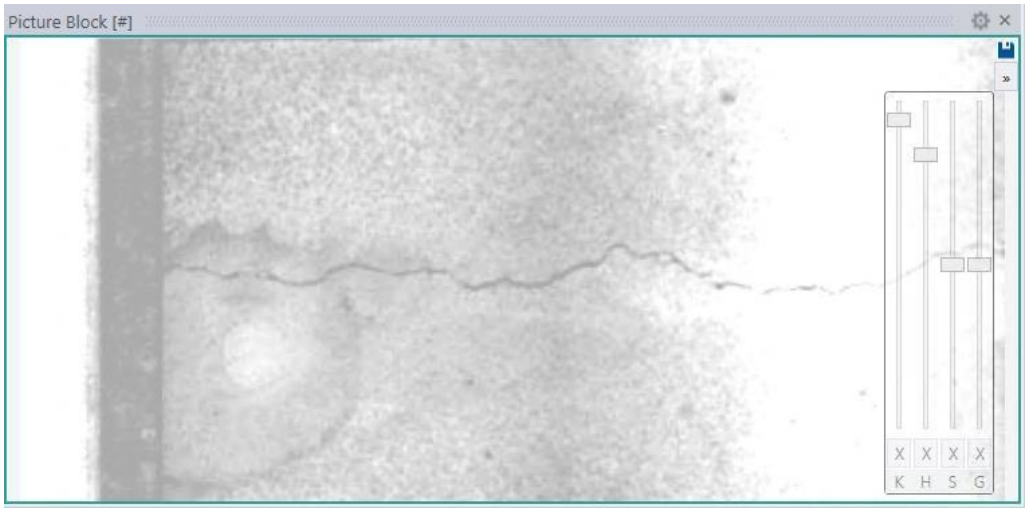


The configuration of this view allows the selection of several image block channels.
Reference to the navigation control: The view shows the image at the current position of the navigation control. If there are several ‘valid’ images due to several selected channels, the one to be displayed is selected according to the following rule:

- Restrict the images that are valid for the currently selected time (i.e. Image.StartTimestamp <= CurrentTimestamp and Image.StartTimestamp + Image.Duration > Current.Timestamp).
- Narrow the images based on the selected image block channels.
- Take the image with the smallest distance between Image.StartTimestamp and CurrentTimestamp.
- If there are more than one, take a random one.

The mouse wheel can be used to determine (enlarge or reduce) the displayed area of the image. If the image is larger than the window, it can be moved as long as the left mouse button is pressed. If the right mouse button is pressed, the image size is reset so that the entire image is displayed. The image can be saved via the save icon (💾). A context menu is displayed where you can select how to save the image:

You can choose whether the image with the filters should be applied or whether the original image should be saved. You can also choose whether the image should be saved on the hard disk or on the clipboard.
The image editing functions can be displayed via the edit icon ( ).
These allow you to manually adjust contrast, brightness, saturation and gamma.

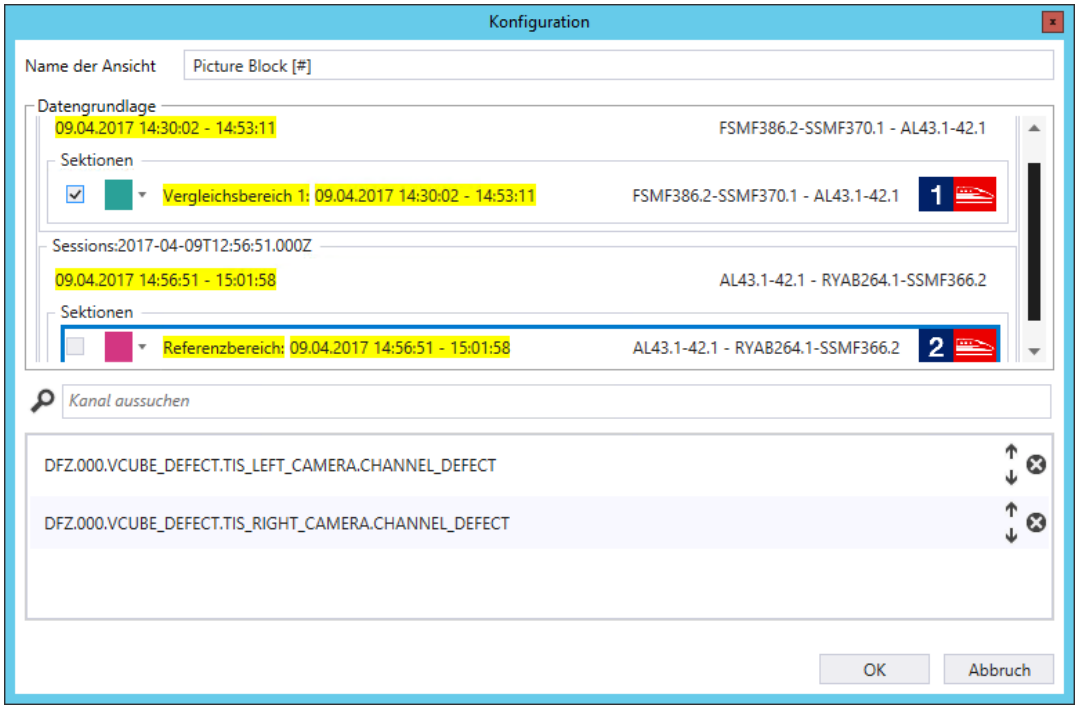


Additional position mode:

- In position mode, a measurement run has to be selected next to the measurement channel for which data should be displayed. The view then receives a frame in the colour of the measurement run as defined in the ‘Loaded Data’ view.

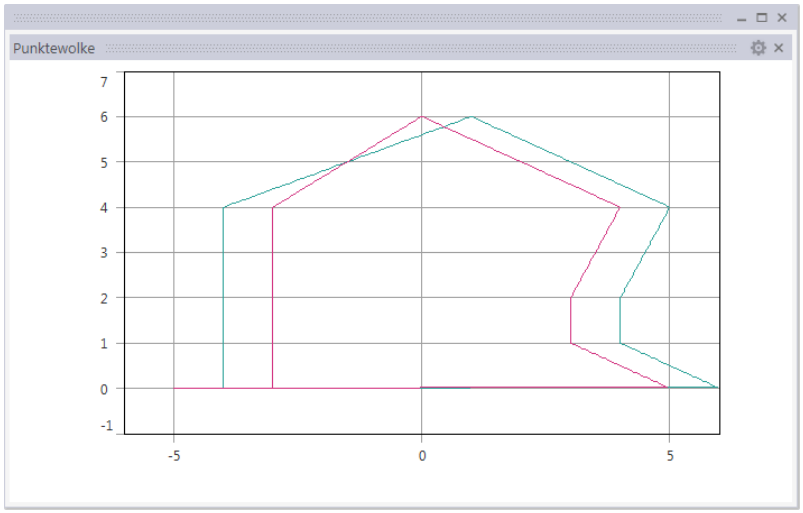
6.7.1. Configuration (Figure)

The following figure shows the configuration dialog in a workspace in position mode:



6.8. View: Point Cloud

The View Point Cloud displays metrics in the ‘Set of 2D Data’ format.



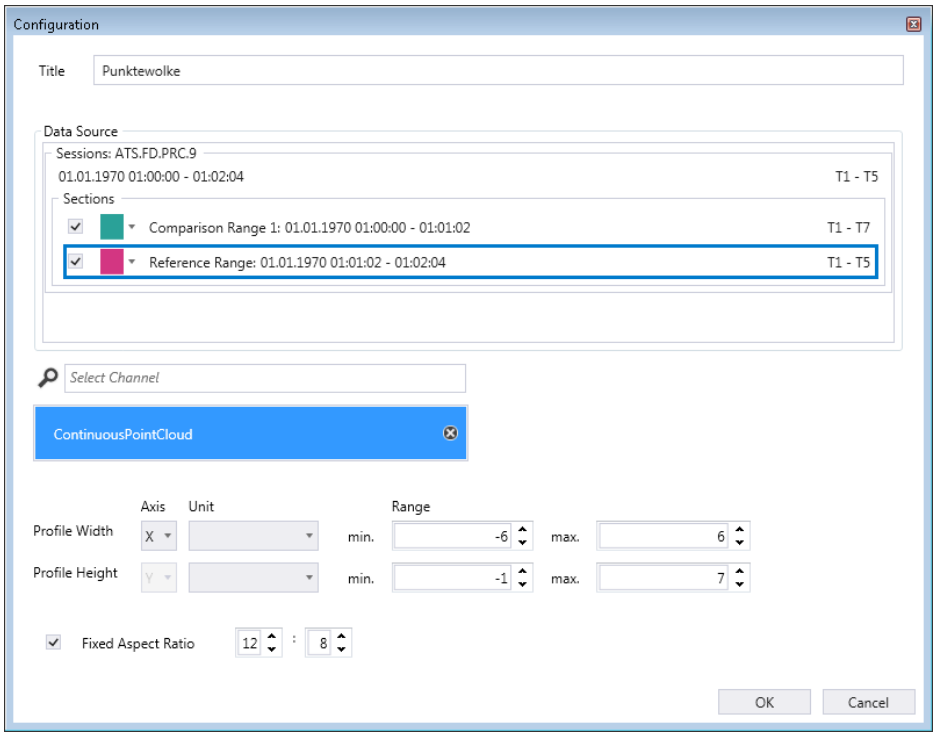
Reference to the navigation control: The view shows the point clouds at the current position of the navigation control.

Difference between time/distance mode and position mode:

- In time/distance mode, a list of measurement channels can be defined in the configuration. The size of the selected measurement channels must match. The view then displays the selected channels as point clouds with different colours.
- In position mode, only one measurement channel can be selected in the configuration. However, it is then possible to select several measurement runs for superposition. The view then shows one point cloud per measurement run. The colour of the point cloud corresponds tot he colour in the ‘Loaded Data’ view. This means that the point clouds of the different measurement runs can be compared.

6.8.1. Configuration (point cloud)

The following figure shows the configuration dialog in a workspace in position mode:



The following settings can be made:

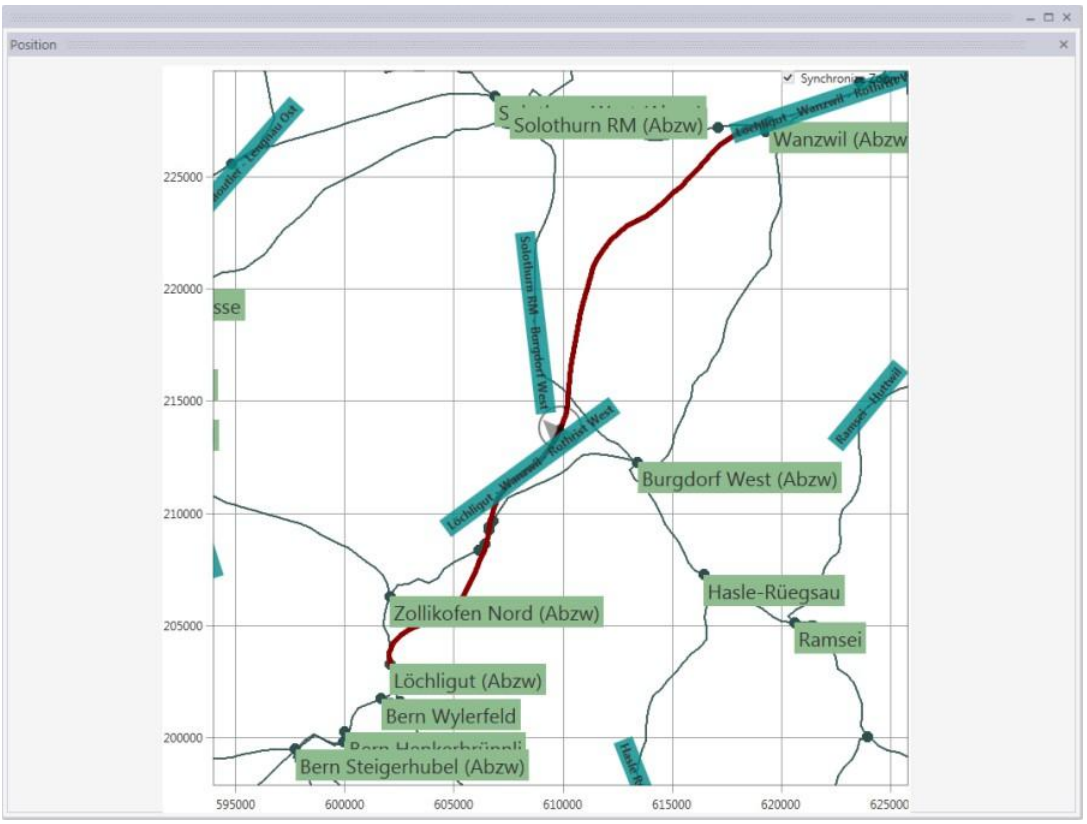
- Units and extent of the x and y axes
- The aspect ratio of the view can be fixed

6.9. View: Position

The view shows all track sections of a track topology in a zoomable view.
Reference to navigation control: The view always shows all elements of the track topology of the reference measurement run (red).

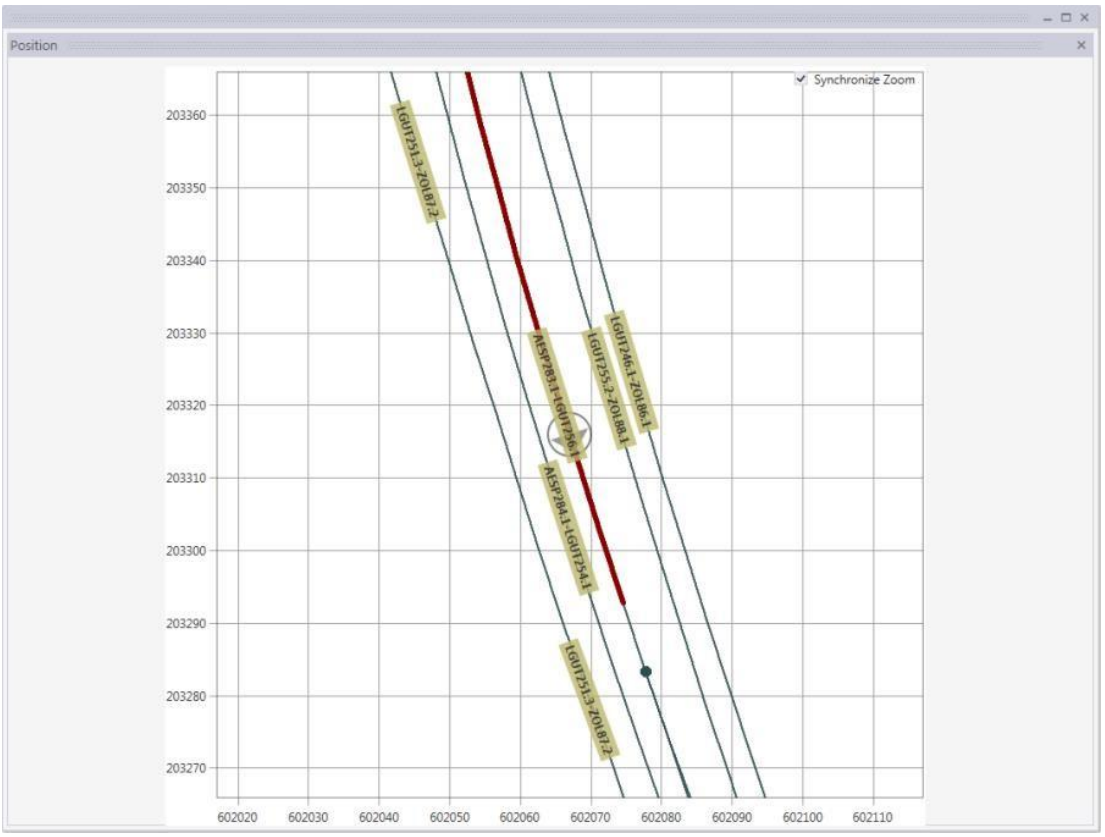
Display of a large map section with lines

When the position view shows a large section of the map, you can see the lines and operating points.



Display of a small map section with track sections

When the position view shows a small section of the map, you can see the track sections.



Presentation of the current reference measurement run

The current reference measurement run is highlighted in colour (red) in the view.

Display of vehicle orientation
Vehicle orientation is indicated by an arrow symbol (grey). The direction of travel is to be understood in the direction of the arrow.

Function: Zooming
You can zoom by scrolling on the mouse wheel.

Function: Pan (moving map section)
The section of the map to be displayed can be panned by holding the two mouse buttons at the same time.

Function: Synchronize Zoom
Using the 'Synchronize Zoom' checkbox, the displayed section can be linked to the visible area in the navigation control.

6.10. View: Position 2.0

View Position 2.0 contains the same functionalities as View Position (see section 6.9 View: Position). However, Position 2.0 View has two limitations. It only works within Switzerland because we can only display SBB maps and an active internet connection is required to load the map material. If a restriction applies, View: Position (section 6.9 View: Position) can be used.

In addition to the existing functions, a map of Switzerland has been added to this view for easier orientation. It is now also possible to use the button to select whether the map or vehicle should always face north when playing a journey (similar to a navigation system in a car).

6.11. View: System Availability Details

The detail view of the system availability consists of two tables. The first table shows all failures and the monitored areas. The second table shows the system messages from the monitoring.
Both tables can be exported as CSV using the disk symbol (💾).

Areas ('summaries')

System Availability Details[#]

Summaries[#] | Messages[#]

Reference[#]	Start-Time[#]	End-Time[#]	Start-Position[#]	End-Position[#]	Level[#]	Message[#]
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:00.633	04.04.2019 11:53:01.452	UZW 21.2 - 34.2 + 132,44	UZW 21.2 - 34.2 + 133,60	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:00.633	04.04.2019 11:53:01.452	UZW 21.2 - 34.2 + 132,44	UZW 21.2 - 34.2 + 133,60	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:00.633	04.04.2019 11:53:01.452	UZW 21.2 - 34.2 + 132,44	UZW 21.2 - 34.2 + 133,60	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:03.809	04.04.2019 11:53:04.926	UZW 21.2 - 34.2 + 138,18	UZW 21.2 - 34.2 + 140,99	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:03.809	04.04.2019 11:53:04.926	UZW 21.2 - 34.2 + 138,18	UZW 21.2 - 34.2 + 140,99	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:03.809	04.04.2019 11:53:04.926	UZW 21.2 - 34.2 + 138,18	UZW 21.2 - 34.2 + 140,99	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:05.455	04.04.2019 11:53:05.944	UZW 21.2 - 34.2 + 142,47	UZW 21.2 - 34.2 + 143,91	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:05.455	04.04.2019 11:53:05.944	UZW 21.2 - 34.2 + 142,47	UZW 21.2 - 34.2 + 143,91	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:05.455	04.04.2019 11:53:05.944	UZW 21.2 - 34.2 + 142,47	UZW 21.2 - 34.2 + 143,91	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:06.400	04.04.2019 11:53:06.488	UZW 21.2 - 34.2 + 145,33	UZW 21.2 - 34.2 + 145,61	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:06.400	04.04.2019 11:53:06.488	UZW 21.2 - 34.2 + 145,33	UZW 21.2 - 34.2 + 145,61	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:06.400	04.04.2019 11:53:06.488	UZW 21.2 - 34.2 + 145,33	UZW 21.2 - 34.2 + 145,61	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:06.661	04.04.2019 11:53:06.996	UZW 21.2 - 34.2 + 146,17	UZW 21.2 - 34.2 + 147,29	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:06.661	04.04.2019 11:53:06.996	UZW 21.2 - 34.2 + 146,17	UZW 21.2 - 34.2 + 147,29	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:06.661	04.04.2019 11:53:06.996	UZW 21.2 - 34.2 + 146,17	UZW 21.2 - 34.2 + 147,29	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:07.396	04.04.2019 11:53:08.147	UZW 21.2 - 34.2 + 148,67	UZW 21.2 - 34.2 + 151,41	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:07.396	04.04.2019 11:53:08.147	UZW 21.2 - 34.2 + 148,67	UZW 21.2 - 34.2 + 151,41	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TMS_	04.04.2019 11:53:07.396	04.04.2019 11:53:08.147	UZW 21.2 - 34.2 + 148,67	UZW 21.2 - 34.2 + 151,41	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_RH_C	04.04.2019 11:53:08.431	04.04.2019 11:53:09.239	UZW 21.2 - 34.2 + 152,49	UZW 21.2 - 34.2 + 155,72	Outage[#]	Data loss occurred from SS 5964
DFZ01.VCUBE.LEFT_TIS_C	04.04.2019 11:53:08.431	04.04.2019 11:53:09.239	UZW 21.2 - 34.2 + 152,49	UZW 21.2 - 34.2 + 155,72	Outage[#]	Data loss occurred from SS 5964

The table has the following columns and shows the failures and the monitored area.

- Reference: The reference refers to a system or a data source.
- Reference in Move-Dir: If the data source has been changed for display due to the direction of travel, the corresponding data source is displayed.
- Start-Time: The start time of the outage or monitoring.
- End-Time: The end time of the outage or monitoring.
- Start position: The position at the start time.
- End position: The position at the end time.
- Start line: The line position at the start time.
- End Line: The line position at the end time.
- Length: The length (in m) of the failure
- Level: The level is either Outage or Monitored Area
- Message: Additional description.

Clicking on a line will navigate to the start time.
Double-clicking on a line will navigate and zoom in so that the entire failure or monitored area can be displayed.

Individual messages

System Availability Details[#]

Summaries[#] | Messages[#]

Reference[#]	Time[#]	Position[#]	Level[#]	Message[#]
DFZ01.VCUBE	04.04.2019 11:52:58.665	UZW 21.2 - 34.2 + 130,55	Info[#]	Connected
DFZ01.VCUBE.RIGHT_RH_C.	04.04.2019 11:52:58.665	UZW 21.2 - 34.2 + 130,55	Info[#]	Connected
DFZ01.VCUBE.RIGHT_TIS_C.	04.04.2019 11:52:58.665	UZW 21.2 - 34.2 + 130,55	Info[#]	Connected
DFZ01.VCUBE.RIGHT_TMS_	04.04.2019 11:52:58.665	UZW 21.2 - 34.2 + 130,55	Info[#]	Connected
DFZ01.VCUBE.CENTRAL_TIS	04.04.2019 11:52:59.039	UZW 21.2 - 34.2 + 130,81	Info[#]	Connected
DFZ01.VCUBE.CENTRAL_TM	04.04.2019 11:52:59.039	UZW 21.2 - 34.2 + 130,81	Info[#]	Connected
DFZ01.VCUBE.CENTRAL_TIS	04.04.2019 11:52:59.039	UZW 21.2 - 34.2 + 130,81	Info[#]	CurrentStatus="Acquisition Stopping"(19)
DFZ01.VCUBE.CENTRAL_TM	04.04.2019 11:52:59.039	UZW 21.2 - 34.2 + 130,81	Info[#]	CurrentStatus="Acquisition Stopping"(19)
DFZ01.VCUBE.LEFT_RH_CAI	04.04.2019 11:52:59.425	UZW 21.2 - 34.2 + 131,13	Info[#]	Connected
DFZ01.VCUBE.LEFT_TIS_CAI	04.04.2019 11:52:59.425	UZW 21.2 - 34.2 + 131,13	Info[#]	Connected
DFZ01.VCUBE.LEFT_TMS_C/	04.04.2019 11:52:59.425	UZW 21.2 - 34.2 + 131,13	Info[#]	Connected
DFZ01.VCUBE	04.04.2019 11:52:59.425	UZW 21.2 - 34.2 + 131,13	Info[#]	Connected
DFZ01.VCUBE.RIGHT_RH_C.	04.04.2019 11:52:59.821	UZW 21.2 - 34.2 + 131,51	Error[#]	FaultDescription="Trigger Check IVP Trigg
DFZ01.VCUBE.RIGHT_TIS_C.	04.04.2019 11:52:59.821	UZW 21.2 - 34.2 + 131,51	Error[#]	FaultDescription="Trigger Check IVP Trigg

The table has five columns and shows the system messages from the monitored system.

- Reference: The reference refers to a system or a datasource.
- Time: The time of the message.
- Position: The position at the time.
- Level: The level of the message.
- Message: Details of the message.

Clicking on a line will navigate to the time of the message.

6.12. View: Tracks and Track Objects Overview

This view shows all tracks of the currently loaded measurement run in a table. NOTE: This view is only available in position mode. No content can be displayed in time/distance mode.

Übersicht über Gleise und Gleis-Objekte

55 Gleise in diesem Abschnitt gefunden

Gleis Name	Typ	Länge	Richtung	GTG Id	Strat Navigationspunkt	Ende Navigationspunkt
BSRB 354.1-3	Weichenstrang	32.58	right	c893a46c-9fb7-11e7-bcdf-d3fb3f0ce5f7	783.94	816.52
BSRB 354.3 - 355.3	Stationsstrang	30.47		25df4f5c-9fb8-11e7-bcdf-d3fb3f0ce5f7	816.52	846.99
BSRB 355.1-3	Weichenstrang	33.20	right	258ba695-9fb7-11e7-bcdf-d3fb3f0ce5f7	846.99	880.19
BSRB 355.1 - 647.1	Stationsstrang	205.40		4b71d6ec-9fb8-11e7-bcdf-d3fb3f0ce5f7	880.19	1085.59
BSRB 647.1-3	Weichenstrang	33.20	right	defa5a20-9fb7-11e7-bcdf-d3fb3f0ce5f7	1085.59	1118.78
BSRB 647.3 - 648.4	Stationsstrang	46.38		df9adff9-9fb7-11e7-bcdf-d3fb3f0ce5f7	1118.78	1165.16
BSRB 648.2-4	Weichenstrang	28.36	right	df0cd0b3-9fb7-11e7-bcdf-d3fb3f0ce5f7	1165.16	1193.52
BSRB 648.2 - 649.1	Stationsstrang	7.62		dfaae58c-9fb7-11e7-bcdf-d3fb3f0ce5f7	1193.52	1201.14

alle verfügbaren Gleis-Objekte nur Objekte auf dem ausgewählten Gleis

1476 Gleis-Objekte werden in der Liste angezeigt

Gleis Name	TrackObject ID	TrackObject Name	Zusätzliche Informationen	Position Start	Position Ende
BSRB 120.2 - 242.2	201	BP: N (Normal)		0	12.371
BSRB 120.2 - 242.2	201	BP: N (Normal)		12.371	127.7508
BSRB 120.2 - 242.2	301	Sleeper Type: Holz	1	0	12.371
BSRB 120.2 - 242.2	301	Sleeper Type: Holz	1	12.371	127.7508
BSRB 120.2 - 242.2	406	Fastening Type: K Wood		0	12.371
BSRB 120.2 - 242.2	406	Fastening Type: K Wood		12.371	127.7508
BSRB 120.2 - 242.2	500	Anchors		0	48.38
BSRB 120.2 - 242.2	600	Straight Track		0	52.569

In the upper part of the view, the tracks are displayed in the order in which they were crossed. By selecting a track, the current position is set to that track. The track with the red border is the one on which the current position is located. Several tracks can be selected using the 'Shift' and 'Ctrl' keys. The zoom is automatically adjusted so that all selected tracks are in the visible area.

The track section objects are displayed in the lower part. Using the radio button between the two tables you can select whether only the objects of the current track section or all are displayed. The track section objects can also be used to set the position by clicking on them. In both tables, it is possible to filter for each column. For example, it is possible to display only dilatations (expansion joints) and level crossings (level crossings).

Gleis Name	TrackObject ID	TrackObject Name	Zusätzliche Informationen	Position Start	Position Ende
AA 9.2 - 40.2	1706	Level crossing	BUe Aarau GB 2	155.499	155.499
DOT 5.2 - HDK 32.1	1706	Level crossing	BUe Ammerswilerstr.	181.086	181.086
HDK 9.2 - 24.1	1706	Level crossing	BUe Hauptstrasse	294.399	294.399
DOT 5.2 - HDK 32.1	1706	Level crossing	BUe Dintikerstrasse	608.9561	608.9561
OI 18.1 - RK 64.1	913	Expansion Joint	65527@725.389	733.8376	716.9396
OL 473.2 - RTR 16.2	913	Expansion Joint	134634@759.803	768.2522	751.3532
DOT 5.2 - HDK 32.1	1706	Level crossing	BUe Kirchweg	1300.172	1300.172
BN7 1.1 - MI 43.2	1706	Level crossing	BUe Mitteldorf	2994.539	2994.539

As already mentioned above, the resulting entries can be accessed by selection. This can be interesting, for example, to see a dilatation in the raw images of the rail.

6.13. View: Value Exceeded (Value Inspection)

This view allows the user to find value exceedances within the loaded data. The corresponding values can be defined by the user. These values can be set in the configuration (see chapter 6.13.1 Configuration (value exceeded)).

Value Inspection

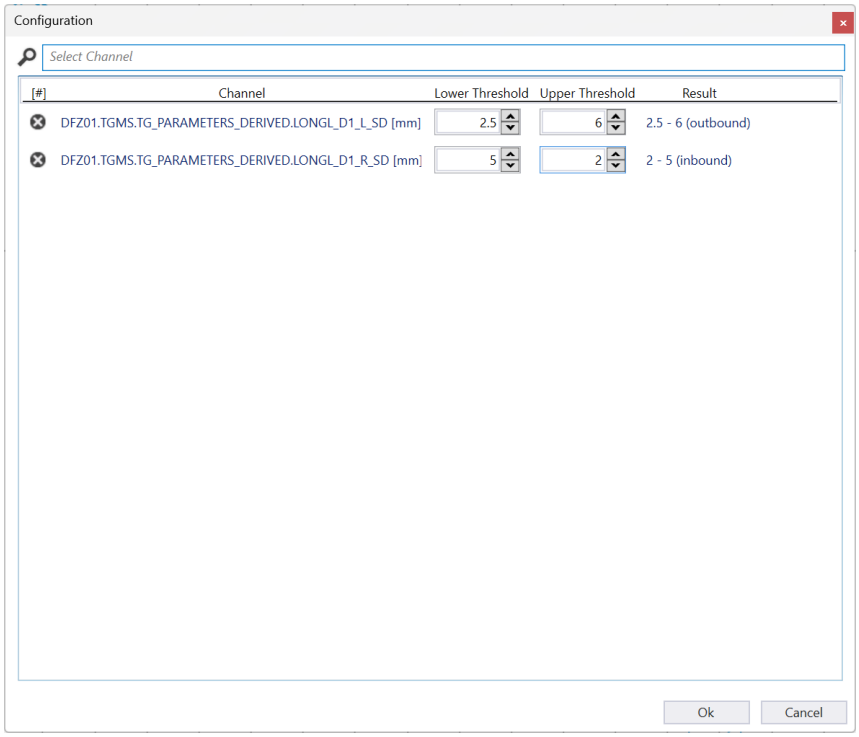
Refresh

Start Position	End Position	Time	Duration	Length [#]	Maximum Violation [	Measurement Channe	Session
BEL 643.1-2 + 18.39	BEL 642.1-2 + 18.45	30.03.2024 13:31:51	00:00:05.0302896	26.25	0 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
BEL 641.1-2 + 10.50	BEL 641.1-2 + 1.01	30.03.2024 13:32:09	00:00:45.3734702	9.5	2.435943365097046 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
BEL 480.1 - 641.1 + 43.91	BEL 480.1 - 641.1 + 24.42	30.03.2024 13:32:55	00:00:04.0579425	19.5	2.389014482498169 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
BEL 480.1-2 + 22.66	BEL 479.1-3 + 15.10	30.03.2024 13:33:06	00:00:05.0341731	37	2.317704439163208 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
BEL 479.1-3 + 17.87	BEL 478.4 - 479.3 + 2.90	30.03.2024 13:33:11	00:00:01.6597890	12.25	2.442016363143921 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
BEL 478.3-4 + 30.23	CASN 163_114N.1 - CAST 1	30.03.2024 13:33:14	00:03:22.3001187	3708	0.49773478507995605 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883
CASN 163_114N.1 - CAST 1	CASN 163_114N.1 - CAST 1	30.03.2024 13:36:36	00:00:00	0	0.09589254856109619 mm	DFZ01.TGMS.TG_PARAMET	20240330_132344.883

The ranges found in the measurement data are displayed as a list (see above). An entry in the list corresponds to the range from when the value is exceeded until the value leaves it again. With a double click, the viewer sets the position and length of the selected area. The values themselves are not displayed in this view. For this, a view must also be displayed 'Values over time / position' can be set up with the corresponding channel.

6.13.1. Configuration (value exceeded)

The following figure shows the configuration dialog of the View Value Overflow:



Values can be entered here for channels that have a unit (e.g. mm, %, °C, etc.). The RCM-DX Viewer uses these values to search for the corresponding ranges with value exceedances in the measurement data and displays them in a list.

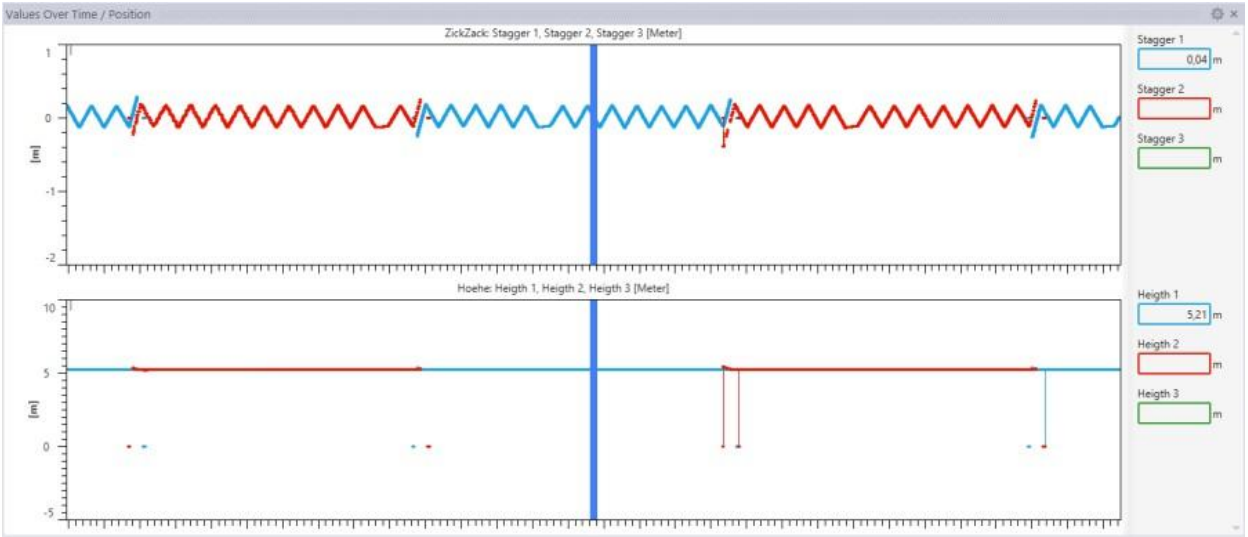
For the value of the exceedance, the lower and upper values must always be set.

Outbound
If the lower value is smaller than the upper value, ranges outside (outbound) of these values are searched.

Inbound
If the lower value is greater than the upper value, ranges within (inbound) these values are searched.

6.14. View: Values Over Time / Position

The ‘Value over time / position’ view displays one-dimensional measured values as a plot on a time or position axis.



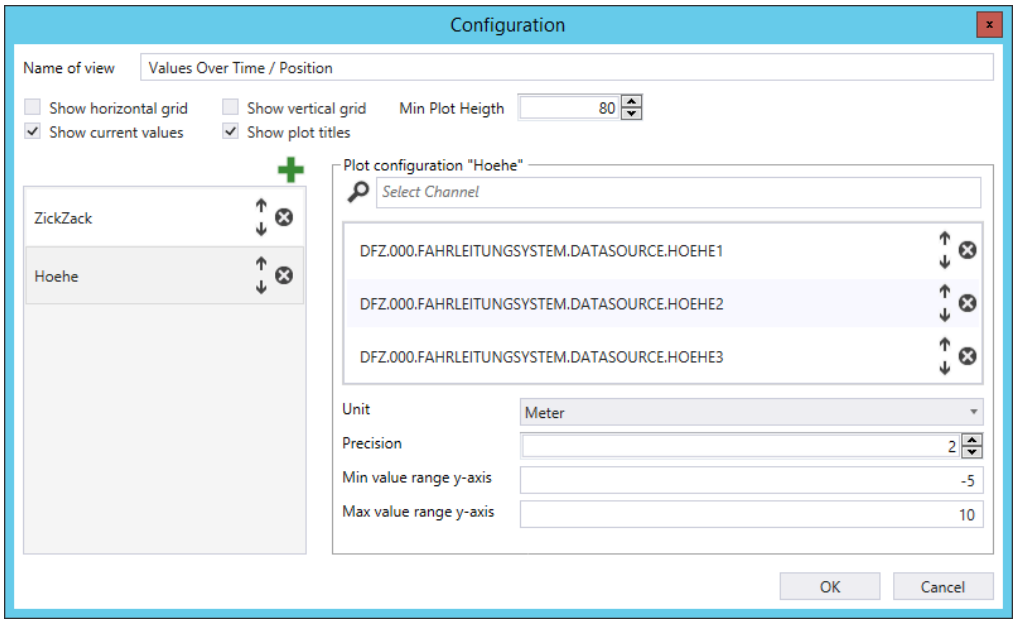
Relation to the navigation control: The view shows curves in the visible area defined by the navigation control.
Limit value violations are highlighted in colour in the plot (background and curve).
Difference between time/distance mode and position mode:

- **In time/distance mode**, a list of plots can be defined in the configuration and a list of measurement channels can be defined for each plot. The size of the selected measurement channels must match per plot. The view then displays the selected channels in each plot as curves with different colours.

- **In position mode**, several measurement runs can be selected in the configuration for overlapping. The view then displays a curve in each plot per measurement run and channel. The colour of the curve corresponds to the colour defined in the Loaded Data view if only one channel has been selected. If several channels have been selected, one colour can be defined per channel. The colour of the frame of the curves corresponds to the colour defined in the Loaded Data view. The curves of the different measurement runs can therefore be compared per plot or measurement channel.

6.14.1. Configuration (values over time/position)

Configuration in time/distance mode
The following figure shows the configuration dialog in a workspace in Time/Distance mode:

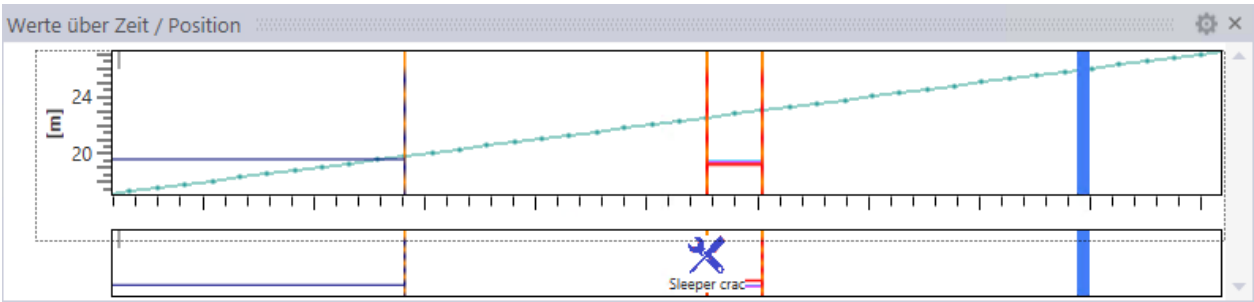


The following can be set for each plot:

- The unit to be displayed
- The number of decimal places for the current values (if displayed)
- The extent of the y-axis. You can also enter 0 for both fields here. The extent then adjusts to the displayed data.
- The plot name can be changed by double-clicking.

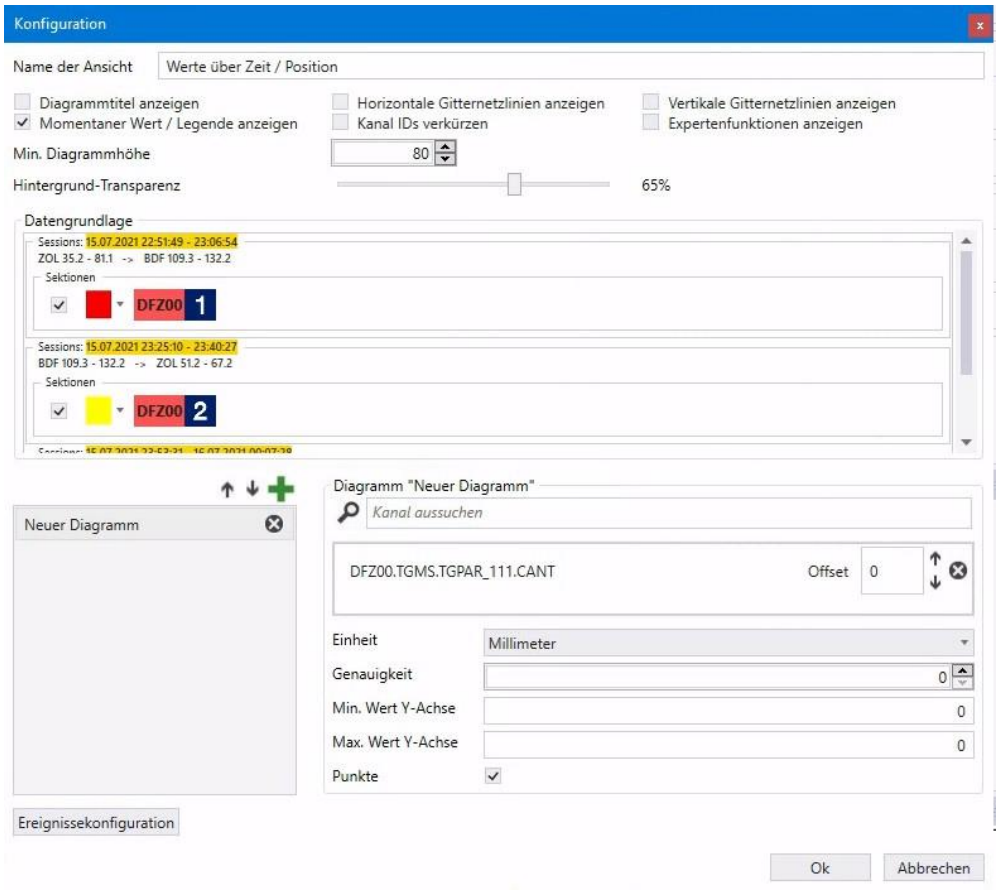
The button ‘Event Configuration’ (not shown above) can be used to adjust the display of events. The functions here are identical to those in the navigation control.

The plot with activated events looks like this:



Configuration in Position mode

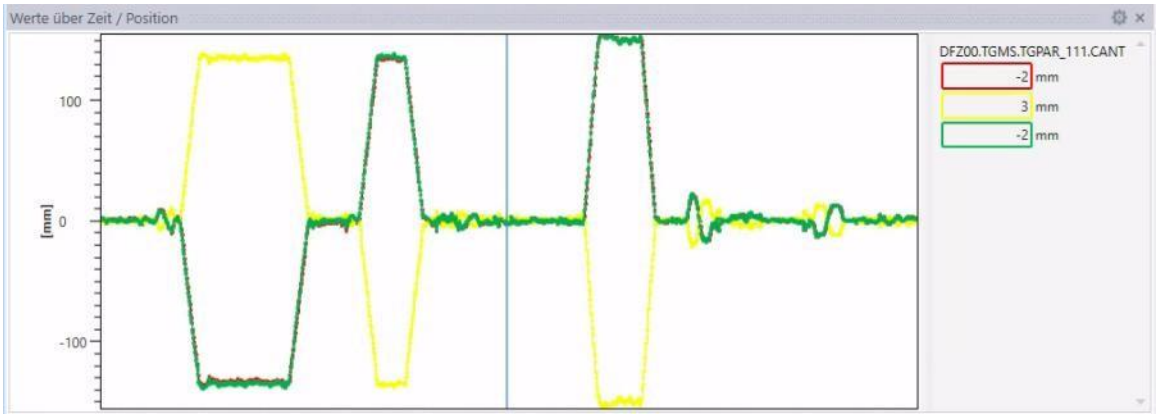
The following figure shows the configuration dialog in a workspace in Position mode:



The following can be set for each plot:

- The unit to be displayed.
- The number of decimal places for the current values (if displayed).
- The extent of the y-axis. You can also enter 0 for both fields here. The extent then adjusts to the displayed data.
- The plot name can be changed by double-clicking.
- One colour can be selected per channel.
- One offset can be defined per channel. All data from this channel is offset in the Y direction.

The plot with the current values looks like this:

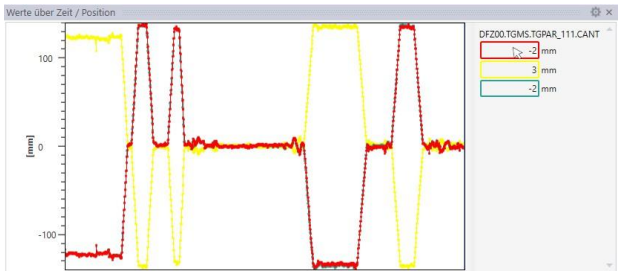


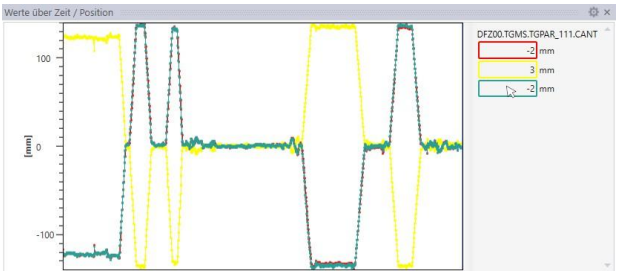
In the current values view, the channel name is displayed with the selected colour. The border of the current values corresponds to the colours defined in the ‘Loaded Data’ view.

If only one measurement run has been selected, the colour defined in the ‘Loaded Data’ view will appear as the border colour.

Highlighting plots

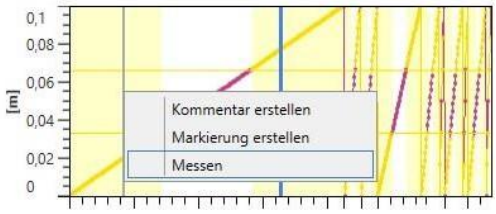
If the mouse points to one of the current values, the corresponding plot is highlighted.





Measuring of areas

In the plot it is possible to measure an area:
Right-clicking on the plot opens a context menu where you can select measurements. This option is only available when you are in position mode.

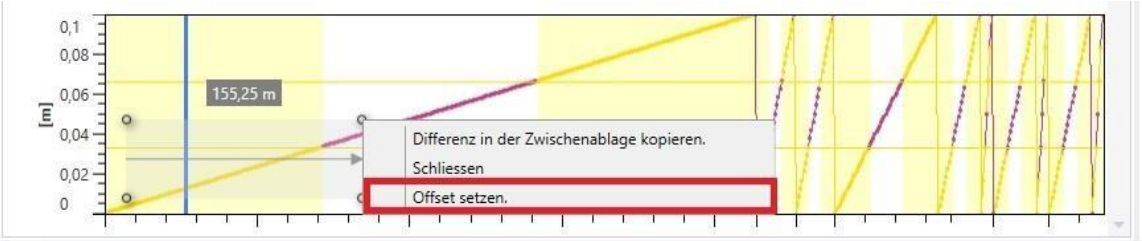


As soon as you press Measure, the mouse display changes and a crosshair appears. The first click starts to define the area to be measured. The second click completes the measurement and displays the result. A context menu also opens automatically, via which the measurement can be easily closed or the measurement result can be copied to the clipboard.

Measuring and setting an offset

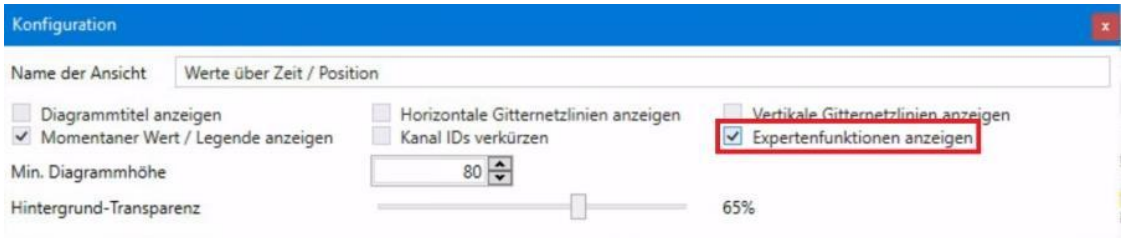
In the ‘Loaded Data’ view, each comparison section has a ‘Measure’ button. This button can be activated and defines the section for which the offset to be measured should apply.

The measurement is then carried out as described above (see 2.10.3.4). However, after the measurement, the option ‘Set offset’ is available in the context menu. If this option is selected, the measured offset is automatically adopted.



Expert functions

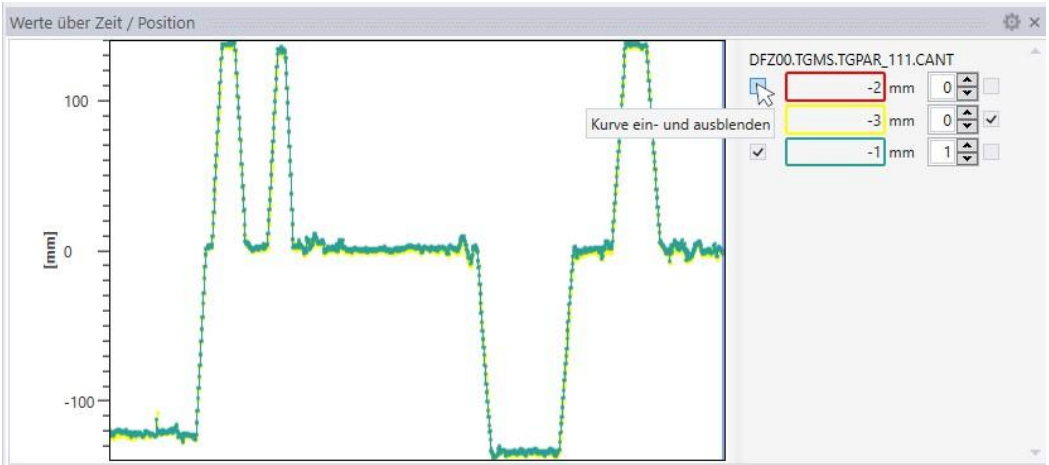
The expert functions offer the option of manipulating each plot individually. They are not available by default. They can be activated via the setting ‘Show expert functions’ in the configuration dialog.



The following functions are available per plot:

- Show and hide the plot: In the example below, the red plot is hidden.
- Apply a Y-offset to the plot: In the example below, the green plot is shifted up by 1mm.
- Inversion of the plot (multiplication of the values by -1): In the example below, the yellow plot is inverted.

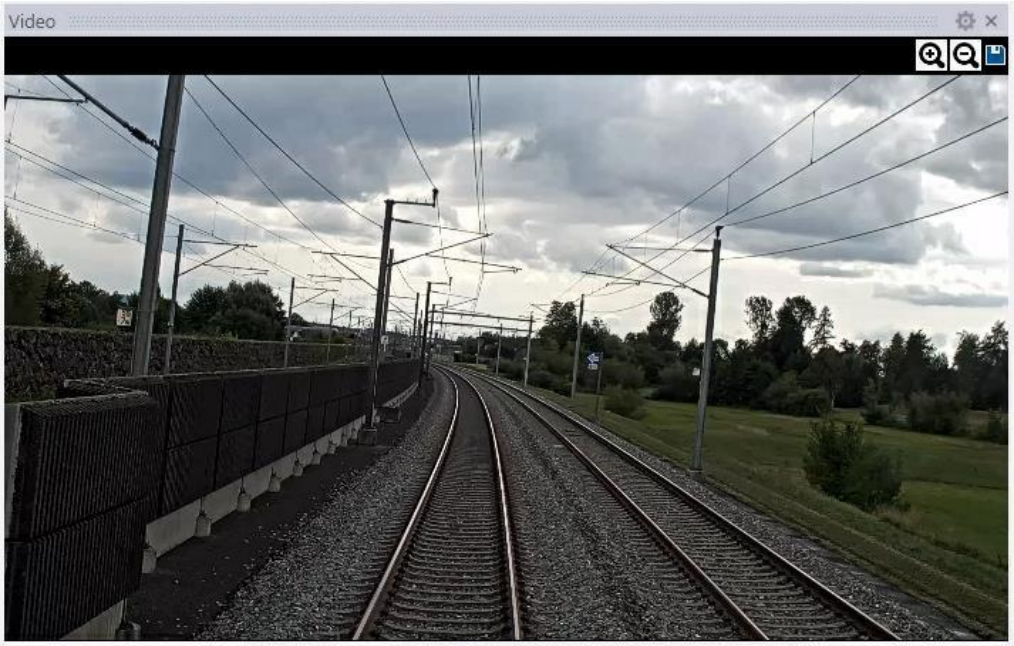
To save space, the control elements are not labelled. If you hold the mouse for a short time, a tooltip explains how they work.



When loading several sections, the inversion of certain channels is automatically activated. These are channels whose sign depends on the direction of travel (these channels were marked with the ‘Move Direction Auto Invert’ flag in the gDFZ and DFZ pipeline).

6.15. View: Video

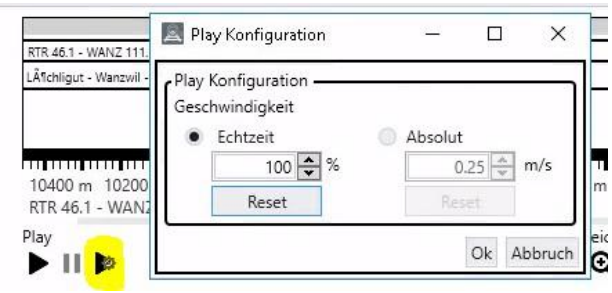
This view represents measurements of the ‘video stream’ type.



Reference to the navigation control: The view shows the video image at the current position of the navigation control. The playback of the video is controlled via the navigation control:

- Play mode: Video plays at recording speed from current position
- Pause mode: Video is paused at current position

The playback speed can be adjusted in relation to the driving speed via the Play-Config button.



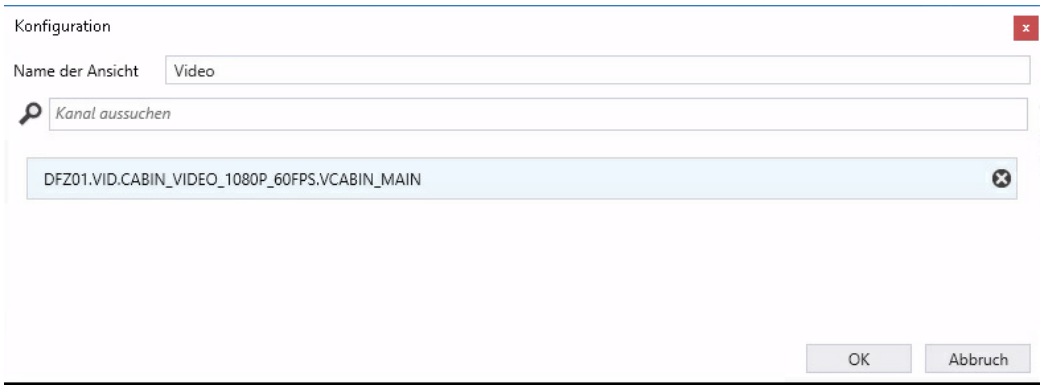
If an absolute playback speed is configured in the above dialog, the video will not be displayed (only in pause mode a still image will appear). This limitation has a technical background and would require permanent adjustment of the video speed.

In pause mode, the zoom icons ( ) can zoom in on a section of the video. The save icon ( ) saves the current video image.
Difference between the Time/Distance Mode and Position Mode:

- There is no difference between the two modes. Only one measurement channel can be selected in the configuration. In position mode, the video of the reference measurement run is always displayed.

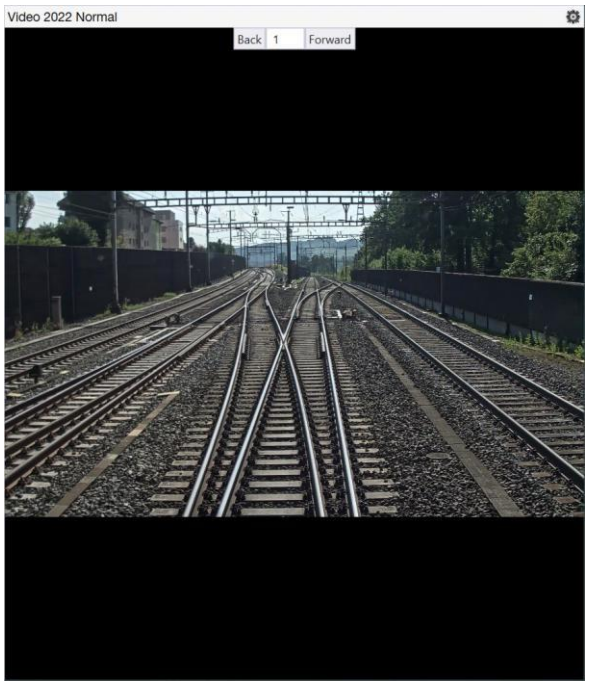
6.15.1. Configuration (video)

The following figure shows the configuration dialog of the View Video:



6.16. View: Video 2022

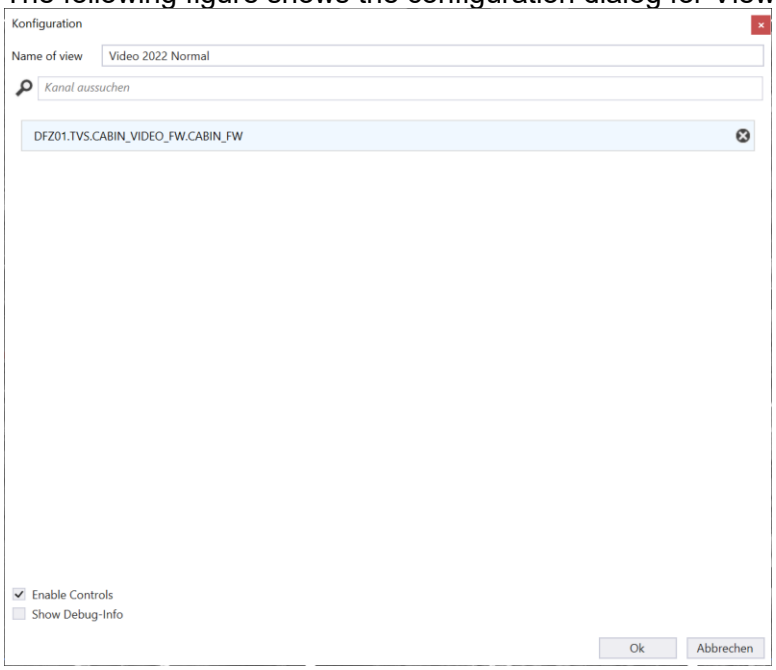
This view has the same functions as the View Video (see Chapter 6.15). However, it also includes a mechanism to move the position by a certain number of frames.

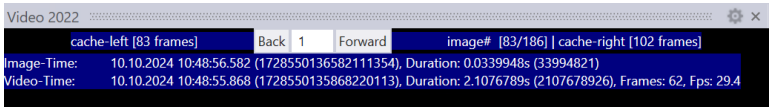


This control element can be switched on and off in the configuration (see chapter 6.16.1 Configuration (video 2022)).
An integer value can be specified in the control element itself, which determines the number of frames to be skipped forwards or backwards when the mouse wheel is pressed. For each tick of the mouse wheel, the frames are skipped and the position in the RCM-DX Viewer is reset accordingly, so that all frames and values match again.

6.16.1. Configuration (video 2022)

The following figure shows the configuration dialog for View Video 2022:



Enable controls
Activating the checkbox activates the control elements for the video display.

Show Debug Info
Activating the checkbox displays the debug information about the video